

# Morphine: Drug information - UpToDate

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Morphine: Drug information

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For additional information see "Morphine: Patient drug information" and "Morphine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: <https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term>

ALERT: US Boxed Warning

Risks with neuroaxial administration (Duramorph, Infumorph, Mitigo):

Single-dose neuraxial administration may result in acute or delayed respiratory depression for up to 24 hours. Because of the risk of severe adverse reactions when administered by the epidural or intrathecal route of administration, patients must be observed in a fully equipped and staffed environment for at least 24 hours after the initial dose.

Ethanol use (extended-release capsules):

Instruct patients not to consume alcoholic beverages or use prescription or nonprescription products that contain alcohol while taking morphine extended-release (ER) capsules. The coingestion of alcohol with morphine ER may result in increased plasma levels and a potentially fatal overdose of morphine.

Addiction, abuse, and misuse:

Because the use of morphine exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing

and reassess all patients regularly for the development of these behaviors and conditions.

Opioid analgesic Risk Evaluation and Mitigation Strategy (REMS) (oral formulations):

Health care providers are strongly encouraged to complete a REMS-compliant education program and to counsel patients and caregivers on serious risks, safe use, and the importance of reading the Medication Guide with each prescription.

Life-threatening respiratory depression:

Serious, life-threatening, or fatal respiratory depression may occur with use of morphine, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of morphine are essential. Instruct patients to swallow morphine ER capsule and tablet formulations whole. May sprinkle contents of the ER capsule on applesauce and swallow immediately without chewing. Crushing, chewing, or dissolving the tablets or contents within the capsule can cause rapid release and absorption of a potentially fatal dose of morphine. Because of delay in maximum CNS effect with IV administration (30 minutes), rapid IV administration may result in overdosing. Observe patients in a fully equipped and staffed environment for at least 24 hours after each test dose of Mitigo and, as indicated, for the first several days after surgery.

Neonatal opioid withdrawal syndrome:

Advise pregnant women using opioids for an extended period of time of the risk of Neonatal Opioid Withdrawal Syndrome, which may be life-threatening if not recognized and treated. Ensure that management by neonatology experts will be available at delivery.

Accidental ingestion (oral and rectal formulations):

Accidental ingestion of even one dose of morphine, especially by children, can result in a fatal overdose of morphine.

Risk of medication errors (oral solution):

Ensure accuracy when prescribing, dispensing, and administering morphine oral solution. Dosing errors due to confusion between mg and mL, and other morphine solutions of different concentrations, can result in accidental overdose and death.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of morphine and benzodiazepines or other CNS depressants for use in patients for whom alternative treatment options are inadequate.

Brand Names: US

- Duramorph;
- Infumorph 200;
- Infumorph 500;
- Mitigo;
- MS Contin

Brand Names: Canada

- BAR-Morphine SR [DSC];
- Doloral;
- Kaadian;
- Kaadian SR;
- M-Ediat;
- M-Eslon;
- Morphine HP;
- Morphine LP Epidural;
- Morphine Sulfate SDZ;
- MS Contin;

- MS/IR;
- PMS-Morphine Sulfate;
- SANDOZ Morphine SR;
- Statex;
- TEVA-Morphine SR

#### Pharmacologic Category

- Analgesic, Opioid

Dosing: Adult

#### **Dosage guidance:**

**Safety:** Consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmefene) for patients with factors associated with an increased risk for overdose, such as history of overdose or substance use disorder, patients with sleep-disordered breathing, higher opioid dosages ( $\geq 50$  morphine milligram equivalents [MME]/day orally), and/or concomitant benzodiazepine or gabapentinoid use (Ref).

**Dosing:** Dosing provided is based on typical doses; some patients may require higher or lower doses. Dosing should be individualized based on patient-specific factors (eg, comorbidities, severity of pain, degree of opioid experience/tolerance) and titrated to patient-specific treatment goals (eg, improvement in function and quality of life, decrease in pain using a validated pain rating scale). Use the lowest effective dose for the shortest period of time.

**Clinical considerations:** Opioids may be part of a comprehensive, multimodal, patient-specific treatment plan for managing moderate to severe pain. Maximize nonopioid analgesia (when appropriate) prior to initiation of opioid analgesia (Ref).

Acute coronary syndrome, refractory ischemic chest pain

#### **Acute coronary syndrome, refractory ischemic chest pain:**

**Note:** Use only in patients with continued ischemic chest pain despite maximally tolerated anti-ischemic medications (Ref). Routine use in patients with acute coronary syndrome has been associated with worse clinical outcomes, and concomitant use with oral P2Y<sub>12</sub> inhibitors may diminish antiplatelet effects (Ref).

**IV:** 2 to 4 mg initially, followed by 2 to 8 mg every 5 to 15 minutes as needed (Ref).

Acute pain in opioid-naïve patients

#### **Acute pain in opioid-naïve patients:**

**General dosing: Note:** For acute non-cancer-related pain severe enough to require an opioid, utilize multimodal pain control, maximize nonopioid analgesics, and limit the quantity prescribed to the expected duration of pain severe enough to require opioids. Long-acting preparations for treatment of acute pain in opioid-naïve patients should generally be avoided (Ref).

**Oral:** Opioid-naïve patients:

*Immediate release:* Oral solution, Tablet:

**Note:** Consider the use of other more commonly prescribed oral opioids (eg, oxycodone) instead of morphine (Ref). The 100 mg/5 mL (or 20 mg/mL) concentrated oral solution is not intended for opioid-naïve patients.

**Initial:** 7.5 to 15 mg every 3 to 4 hours as needed; if pain is not relieved, may increase dose as tolerated. Usual dose range: 7.5 to 15 mg every 3 to 4 hours as needed. May give up to 30 mg every 4 hours as needed for severe, acute pain in hospitalized patients at low risk for respiratory depression (Ref).

**IV:** Opioid-naïve patients:

*Intermittent:* Initial: 1 to 4 mg every 1 to 4 hours as needed; if pain is not relieved, may increase dose as tolerated. May give up to 10 mg every 4 hours as needed for severe, acute pain in hospitalized patients at low risk for respiratory depression (Ref). For some severe acute pain episodes (eg, trauma), may initially give more frequently (eg, every 5 to 15 minutes) if needed and titrate to pain relief; once pain relief is achieved, reduce frequency (eg, every 3 to 4 hours as needed) (Ref).

*Patient-controlled analgesia (Ref) :*

**Morphine: Example PCA Initial Dose Ranges for Opioid-Naive Patients<sup>a</sup>**

<sup>a</sup> PCA = patient-controlled analgesia; for use to maintain pain control after initial pain control achieved. If analgesia is inadequate, may adjust dosing and provide rescue bolus doses (Buys 2025).

<sup>b</sup> Due to the risk of respiratory depression, use of a basal dose should be limited to carefully selected patients who are opioid tolerant and/or receiving care in a critical care unit (Buys 2025).

Usual concentration

1 mg/mL

Demand dose

Usual range: 0.5 to 2 mg

Basal dose

In general, a continuous (basal) infusion is **not** recommended in an opioid-naive patient (ISMP 2009)<sup>b</sup>

Lockout interval

6 to 10 minutes

**IM (not recommended for routine use):** Opioid-naive patients: Initial: 5 to 10 mg every 3 to 4 hours as needed; if pain is not relieved, may increase dose as tolerated. **Note:** IM administration is generally not recommended due to pain associated with injection, variable absorption, and delayed time to peak effect (Ref).

**Rectal** (may be used as an alternative to IV or oral administration): Opioid-naive patients: Initial: 10 mg every 4 hours scheduled or as needed; may increase or decrease the dose as tolerated following the same precautions as oral dosing up to 30 mg every 4 hours scheduled or as needed.

Acute pain in opioid-tolerant patients

**Acute pain in opioid-tolerant patients (eg, breakthrough cancer pain):**

**Oral, IV, SUBQ:**

*Intermittent:* Usual dose: In conjunction with the scheduled long-acting opioid, administer 5% to 20% of the basal daily morphine milligram equivalents (MME) requirement given as needed using an IR formulation with subsequent dosage adjustments based upon response (Ref).

*Patient-controlled analgesia (Ref) :*

**Morphine: Example PCA Initial Dose Ranges for Opioid-Tolerant Patients<sup>a,d</sup>**

<sup>a</sup> PCA = patient-controlled analgesia; MME = morphine milligram equivalents; for use to maintain pain control after initial pain control achieved. If analgesia is inadequate, may adjust dosing and provide rescue bolus doses (Buys 2025).

<sup>b</sup> If patient is chronically taking high doses of opioids (eg, >50 MME per day) or medication for opioid use disorder (ie, methadone or buprenorphine), may initiate with demand doses at the high end of the range and a lockout interval at the low end of the range, while continuing their baseline opioid; monitor such patients closely for pain control and opioid-related side effects and adjust the PCA settings accordingly (Buys 2025).

<sup>c</sup> Due to the risk of respiratory depression, use of a basal dose should be limited to carefully selected patients who are opioid tolerant and/or receiving care in a critical care unit (Buys 2025).

<sup>d</sup> ISMP 2009.

Usual concentration

1 mg/mL

Demand dose<sup>b</sup>

Usual range: 0.5 to 2 mg

Basal dose

In general, a continuous (basal) infusion is **not** recommended in an opioid-naïve patient.<sup>c,d</sup>

Lockout interval<sup>b</sup>

6 to 10 minutes

Acute postoperative pain, post-anesthesia care unit

**Acute postoperative pain , post-anesthesia care unit:**

**Note:** Optimize multimodal perioperative pain management (eg, regional or local anesthesia, nerve blocks, nonopioid analgesics, and other adjuvants) to minimize opioid use (Ref). Refer to institutional protocols.

**IV:** 1 to 3 mg given as frequently as every 5 minutes until adequate pain relief or adverse effects occur (eg, respiratory depression, oxygen desaturation, hypotension). Once initial pain control is achieved, reduce frequency and adjust dose based on opioid tolerance (see the "**Acute pain in opioid-naïve patients**" or "**Acute pain in opioid-tolerant patients**" indication) (Ref).

Acute vaso-occlusive pain in sickle cell disease

**Acute vaso-occlusive pain in sickle cell disease:**

**Note:** Dosing presented is for patients who do not have a pain treatment plan on file and whose opioid dose for prior episodes is unknown. If opioid dose given for a prior episode is known, choose initial dose based on intensity of pain in comparison with previous episode and previous effective dose (Ref). Refer to the individual's personal pain protocol or institutional pain protocol for dosing if available.

**IV:** Initial: 3 to 7 mg (maximum initial dose: 10 mg) given once as soon as possible (eg, within 30 to 60 minutes of presentation); if severe pain continues, dose may be repeated or given at 50% of the original dose every ~20 minutes; may increase dose by up to 25%. Monitor for sedation/respiratory depression (Ref).

Chronic pain, including chronic cancer pain

**Chronic pain, including chronic cancer pain:**

**Note:** Before starting opioid therapy for chronic pain, establish realistic goals for pain and function, and consider how therapy will be discontinued if benefits do not outweigh risks. Opioids, including morphine, are **not** the preferred therapy for chronic noncancer pain due to insufficient evidence of benefit and risk of serious harm; nonpharmacologic treatment and nonopioid analgesics are preferred, with the exception of chronic pain from sickle cell disease and end-of-life care. Consider opioids, including morphine, **only** in patients who experience clinically meaningful improvement in pain and function that outweighs patient safety risks (Ref).

*Opioid-naïve patients:* For noncancer pain, morphine requirement should be established using IR formulations (Ref). In patients with cancer pain, may switch to a long-acting formulation earlier in the course of therapy (Ref).

**Oral: Immediate release:** Oral solution, tablet: **Note:** The 100 mg/5 mL (or 20 mg/mL) concentrated oral solution is not intended for opioid-naïve patients.

*Noncancer or cancer pain:* Initial: 5 to 15 mg every 4 hours as needed or scheduled around the clock for some patients (eg, cancer pain) (Ref). For chronic noncancer pain, most patients will have pain control with initial doses <50 mg/day (Ref).

**IV, SUBQ: Note:** Typically reserved for acute exacerbations or those who cannot tolerate oral administration. For progressive illnesses (eg, cancer), a continuous IV or SUBQ infusion, with or without a patient-controlled analgesia option, can also be used as pain requirements increase. In general, SUBQ dose is equivalent to IV dose (Ref). Individualize dose based on patient's previous opioid intake and appropriate opioid analgesic equivalents; titrate further, if needed, based on level of pain.

*Noncancer or cancer pain: IV:* Initial: 2 to 5 mg every 2 to 4 hours as needed (Ref).

*Cancer pain or palliative care: SUBQ:* Initial: 2 to 5 mg every 3 to 4 hours as needed (Ref). If a continuous SUBQ infusion is employed, refer to institutional protocols; reported dosing varies greatly and is based on practice and patient needs (Ref).

### **Titration:**

**Noncancer pain:** Adjust dose according to patient response; if needed, increase the dose slowly in increments of no more than 25% to 50% of the total daily dose (Ref). **Note:** Dosages  $\geq 50$  MME/day are likely to not have increased benefits in pain relief or function relative to overall risks. To reduce risk of overdose in noncancer pain (excluding patients with sickle cell disease and palliative care), readdress pain and reassess evidence of individual benefits and risks when increasing opioid dosage to  $\geq 50$  MME/day (Ref).

**Cancer pain:** Adjust dose according to patient response; if needed, increase the fixed scheduled dose by 30% to 100% of the total dose taken in the prior 24-hour period including total amount of rescue medication used; if pain score decreased and functional assessment improved, continue current effective dosing (Ref).

*Opioid-tolerant patients (also refer to the section **Dose conversions for pain management**):*

### **Oral: Extended release:**

**Note:** Although manufacturer's labeling contains directions for initiating ER morphine products in opioid-naïve patients with chronic pain, these preparations should not be used as initial therapy. Instead, treatment should be initiated with an IR preparation to more accurately determine the daily opioid requirement and decrease the risk of overdose. Unless pain is associated with cancer, palliative care, or sickle cell disease, reserve ER opioids for patients who have received IR opioids daily for  $\geq 1$  week yet continue to experience severe, continuous pain (Ref).

*Capsules, extended release (once- or twice-daily formulation [polymer coated pellets]):* See **Dose conversions for pain management:** Calculated dose may be administered once daily or in 2 equally divided doses administered every 12 hours; may consider dose reduction with first several doses when converting from IR formulations. Example initial dose: 30 mg once daily or 15 mg every 12 hours. Dose adjustments may be made as frequently as every 1 to 2 days. **Note:** For generic ER capsules, some products are unique with varying dosing, schedule, and titration recommendations; refer to product labeling for specific ER capsule product information.

*Tablets, extended release (MS Contin):* See **Dose conversions for pain management:** Calculated dose may be administered in 2 equally divided doses (every 12 hours) or 3 equally divided doses (every 8 hours); may consider dose reduction with first several doses when converting from IR formulations. Example initial dose: 15 mg every 8 or 12 hours. Dose adjustments may be made as frequently as every 1 to 2 days.

**Dose conversions for pain management: Note:** Equianalgesic conversions serve only as a general guide to estimate opioid dose equivalents. Multiple factors must be considered for safely individualizing conversion of opioid analgesia. In general, for noncancer pain, the decision to convert from an IR to an ER formulation should be individualized and reserved for those with severe continuous pain who have been taking opioids for  $\geq 1$  week (Ref).

*Converting from oral morphine to parenteral morphine:*

Approximate equivalency: 30 mg (**oral** morphine): 10 mg ( **IV/SUBQ** morphine) (Ref).

*Converting from oral IR morphine to oral ER morphine preparations:*

**Capsules (once-daily formulation [pellets]):** Total daily oral morphine dose administered once daily; should not be given more frequently than every 24 hours.

**Capsules (once- or twice-daily formulation [polymer coated pellets]):** Total daily oral morphine dose administered once daily; in patients experiencing inadequate analgesia with once-daily dosing, total daily dose can be administered in 2 divided doses (every 12 hours).

**Tablets (MS Contin):** Total daily oral morphine dose may be administered either in 2 divided doses (every 12 hours) **or** in 3 divided doses (every 8 hours).

*Converting from oral morphine to rectal morphine:*

Although the bioavailability of rectal morphine is believed to approximate oral morphine (ie, 1:1), absorption is variable and may be higher or lower than expected. Therefore, when switching from oral to rectal dosing, a reduction in rectal dose may be necessary (Ref).

*Converting to/from morphine (parenteral or oral) to/from a different opioid (parenteral or oral):*

Refer to published equianalgesic opioid conversion data for guidance (or refer to institutional protocols). Provided conversion ratios are only approximations and substantial interpatient variability exists; therefore, it is safer to underestimate a patient's daily oral requirement and provide breakthrough pain relief with IR formulations rather than risk overestimating daily requirements. **When switching to a new opioid (except to/from methadone), reduce the initial daily calculated equianalgesic dose of the new opioid by 25% to 50% to adjust for lack of complete mu receptor cross-tolerance** (conversions to/from methadone are highly variable and require extreme caution) (Ref).

#### **Discontinuation or tapering of therapy:**

When reducing the dose, discontinuing, or tapering long-term opioid therapy, the dose should be gradually tapered. An optimal tapering schedule has not been established. Individualize tapering based on discussions with patient to minimize withdrawal, while considering patient-specific goals and concerns and the opioid's pharmacokinetics. Proposed initial schedules range from slow (eg, 10% reduction per week or 10% reduction per month depending on duration of long-term therapy) to rapid (eg, 25% to 50% reduction every few days)(Ref). Slower tapers may be appropriate after long-term use (eg, >1 year), whereas more rapid tapers may be appropriate in patients experiencing severe adverse effects. During tapering, patients may be at an increased risk of overdose if they return to their original (or higher) opioid dose or use illicit opioids, due to rapid loss of tolerance; consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmeferine). Monitor carefully for signs/symptoms of withdrawal. If the patient displays withdrawal symptoms, consider slowing the taper schedule; alterations may include increasing the interval between dose reductions, decreasing amount of daily dose reduction, pausing the taper and restarting when the patient is ready, and/or coadministration of an alpha-2 agonist (eg, clonidine) to blunt autonomic withdrawal symptoms and other adjunctive agents to treat GI symptoms and muscle spasms, as needed. Continue to offer nonopioid analgesics as needed for pain management during the taper (Ref).

Cough in select patients

#### **Cough in select patients (alternative agent) (off-label use):**

**Note:** Reserve for patients with unexplained chronic cough lasting >3 months who do not benefit from nonpharmacologic interventions, empiric treatment of common etiologies of chronic cough, and nonspecific antitussive therapy (eg, dextromethorphan, benzonatate) or for moderate to severe cough in the palliative care setting (Ref). When used outside of the palliative care setting, some experts recommend reserving for time-limited trials (eg, 4 to 6 weeks) (Ref). Discontinue therapy if clinical improvement is not observed after 1 to 2 weeks of therapy (Ref).

#### **Extended release:**

*Capsules, extended release (once- or twice-twice daily formulation [polymer coated pellets]):* Initial: **Oral:** 10 mg/day; may increase to 10 mg twice daily or 20 mg once daily based on response and tolerability after  $\geq 1$  week (Ref).

*Tablets, extended release:* Initial: **Oral:** 15 mg once daily (based on timing of maximal symptoms); may increase to 15 mg twice daily based on response and tolerability after  $\geq 1$  week (Ref).

### **Immediate release:**

*Oral solution:* Initial: **Oral:** 5 mg every 4 hours as needed for control of symptoms. **Note:** Some experts reserve use of this regimen for opioid-naïve patients in the palliative care setting (Ref).

Dyspnea in palliative care patients

### **Dyspnea in palliative care patients (off-label use):**

*Opioid-naïve patients:*

#### **Moderate to severe dyspnea (eg, on walking a short distance):**

**Oral, sublingual:** Immediate release (may use 100 mg/5 mL [20 mg/mL] solution): Initial: 2 mg every 2 hours as needed **or** 5 mg every 4 hours **with** 2.5 mg every 2 hours as needed or on an “offer, may refuse” basis (most patients will not need every dose) (Ref).

**SUBQ:** Initial: 2 mg every 2 hours as needed or on an “offer, may refuse” basis (most patients will not need every dose) (Ref).

**Severe dyspnea (eg, at rest or with minimal exertion): SUBQ, IV:** Initial: 2.5 mg; if dyspnea persists and initial dose is well tolerated, may repeat every 30 to 60 minutes (SubQ) or every 15 to 30 minutes (IV). If 2 doses are well tolerated but fail to reduce dyspnea adequately, the dose may be doubled (Ref).

*Opioid-tolerant patients:* **Note:** Higher initial doses will likely be needed.

**Moderate or severe dyspnea (eg, on walking a short distance):** Immediate release: **Oral:** Give 10% to 15% of the basal daily opioid requirement (calculated in morphine equivalents) every 2 hours as needed or on an “offer, may refuse” basis. Consider increasing the regular daily dose by ~25% taking into consideration breakthrough doses used in the previous 24 hours (Ref).

**Severe dyspnea (eg, at rest or with minimal exertion): SUBQ, IV:** Give 5% of the oral basal daily opioid requirement (calculated in morphine equivalents) every 1 hour as needed or on an “offer, may refuse” basis. For breakthrough dyspnea, if already taking a parenteral opioid, may give ~10% of the current parenteral opioid daily dose (Ref).

Neuraxial analgesia

### **Neuraxial analgesia:**

#### **Epidural:**

**Note:** Reserve use for severe pain (eg, after surgery, cancer pain). Must be administered by health care providers skilled in the care of patients receiving intraspinal opioids. Use a **preservative-free** formulation intended for neuraxial use. Example regimens and usual doses are provided; specific regimens and doses vary based on type of surgery, comorbidities, and institutional policies and practices.

**Single dose (using 0.5 or 1 mg/mL preservative-free solution):** Opioid-naïve patients: Usual range: 1.5 to 3.75 mg (may depend upon patient comorbidities) (Ref).

**Continuous infusion (using 0.5 or 1 mg/mL preservative-free solution):** Opioid-naïve patients: Morphine is typically added to a local anesthetic at a usual morphine concentration of 20 to 40 **mcg/mL**; rate of infusion (eg, 3 to 14 mL/hour) varies dependent upon pain management needs, mode of delivery (eg, patient-controlled epidural analgesia), or spinal level of epidural catheter. Lower concentrations of 5 to 10 **mcg/mL** may be used in frail patients or those at risk for post-operative respiratory depression (Ref).

**Continuous microinfusion** (using a device intended for continuous microinfusion): **Note:** Must use a 10 or 25 mg/mL **preservative-free** solution (eg, Infumorph); dilution may be required.

Initial: 3 to 7.5 mg over 24 hours.

#### **Intrathecal:**

**Note:** Reserve use for severe pain (eg, after surgery, cancer pain). Must be administered by health care providers skilled in the care of patients receiving intraspinal opioids. Use a **preservative-free** formulation intended for neuraxial use. Intrathecal dose is usually one-tenth that of epidural dosage.

**Single dose** (using 0.5 or 1 mg/mL **preservative-free** solution): Usual range: 0.075 to 0.3 mg coadministered with a local anesthetic; repeat doses are not recommended. If pain recurs, use of an alternative route of administration is recommended (Ref). **Note:** Doses vary depending on the clinical scenario and other medications administered as part of multimodal analgesia. Risk of adverse effects (eg, nausea, pruritus, respiratory depression) is higher with doses >0.3 mg; however, some patients with chronic intractable pain (eg, cancer pain) may require doses up to 0.5 mg (Ref).

**Continuous microinfusion** (using a device intended for continuous microinfusion): **Note:** Must use a 10 or 25 mg/mL **preservative-free** solution (eg, Infumorph); dilution may be required.

Initial: 0.2 to 1 mg over 24 hours.

Pain and sedation; critically ill patients in the ICU

#### **Pain and sedation; critically ill patients in the ICU (off-label use):**

**Note:** Multimodal approaches (eg, a combination of analgesics and techniques) should typically be employed for pain control in this setting. Pain should be monitored using validated scales (eg, behavioral pain scale, critical-care observation tool) in medical, postoperative, or trauma (excluding brain injury) ICU patients who are unable to self-report (Ref).

#### **IV:**

**Loading dose:** 2 to 10 mg, followed by maintenance dosing (Ref). **Note:** More than 1 loading dose may be needed; onset of action following IV administration is 5 to 10 minutes. Reduce or omit initial loading dose in select patients (eg, older, hypovolemic, at-risk for hemodynamic compromise) (Ref).

**Maintenance dosing:** 2 to 4 mg every 1 to 2 hours or 4 to 8 mg every 3 to 4 hours (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

**Note:** Morphine clearance is similar in patients with altered and normal kidney function. However, glucuronide metabolites (M3G [inactive as an analgesic; may contribute to CNS stimulation and lower seizure threshold] and M6G [active analgesic]) significantly accumulate in patients with reduced kidney function (Ref). As kidney function deteriorates, M6G accounts for an increasing proportion of the analgesic effect and attainment of steady state is delayed; consequently, patients may experience CNS depression, sedation, and severe and prolonged respiratory depression, which may be delayed in presentation (Ref).

#### **Altered kidney function:**

##### **Oral, IM, IV, SUBQ, Rectal:**

**Note:** There are no specific dose adjustments provided in the manufacturer's labeling. ER formulations should preferably be avoided in patients with altered kidney function (Ref). The following suggestions are based on expert opinion and selected references (Ref):

CrCl  $\geq$  60 mL/minute: No dosage adjustment necessary.

CrCl 30 to <60 mL/minute: Consider use of an alternative opioid analgesic. If necessary, administer 50% to 75% of usual initial dose; may also consider extending dose interval. Titrate cautiously to response.

CrCl 15 to <30 mL/minute: Avoid use. If necessary, administer 25% to 50% of usual initial dose; may also consider extending dose interval. Titrate cautiously to response.

CrCl <15 mL/minute: Avoid use.

### **Epidural/intrathecal:**

Moderate to severe impairment: There are no specific dose adjustments recommended; however, use with caution and monitor closely. Use of an alternative opioid analgesic may be preferred in patients with more severe impairment.

**Hemodialysis, intermittent (thrice weekly):** Morphine and M6G are dialyzable, although the extent of removal is not fully quantified (Ref); slow distribution from CNS to plasma may result in prolonged CNS depression even after reduction in plasma levels following dialysis (Ref).

### **Oral, IM, IV, SUBQ, Rectal:**

**Note:** There are no specific dose adjustments provided in the manufacturer's labeling. ER formulations should preferably be avoided in patients with altered kidney function (Ref). The following suggestions are based on expert opinion and selected references (Ref):

Avoid use; consider alternative opioid analgesic. If necessary, administer 25% to 50% of usual dose; may also consider extending dose interval. Titrate cautiously to response.

**Epidural/intrathecal:** There are no specific dose adjustments recommended; however, use with caution and monitor closely. Use of an alternative analgesic may be preferred.

**Peritoneal dialysis:** Avoid use; consider alternative opioid analgesic (Ref). Morphine, M3G, and M6G are not significantly removed by peritoneal dialysis (Ref).

**CRRT:** Other opioids are preferred (eg, hydromorphone, fentanyl) due to more reliable pharmacokinetic profiles (Ref). Morphine is not significantly removed by hemofiltration or hemodiafiltration (Ref), and the extent of M3G and M6G removal is unknown.

**PIRRT (eg, sustained, low-efficiency diafiltration):** Other opioids are preferred (eg, hydromorphone, fentanyl) due to more reliable pharmacokinetic profiles. The extent of morphine, M3G, and M6G removal is unknown (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based up on the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

**Note:** ER formulations should be avoided in patients with liver impairment (Ref). Ensure appropriate prophylaxis against constipation and hepatic encephalopathy (eg, lactulose) (Ref). Monitor for oversedation and/or worsening encephalopathy during use and following dose titration (Ref).

In patients with cirrhosis, morphine AUC and elimination half-life are approximately doubled due to reduced glucuronidation during first pass metabolism that occurs from decreased liver blood flow secondary to portosystemic shunting; additionally, morphine-3-glucuronide (neurotoxic metabolite) and morphine-6-glucuronide (active analgesic) accumulate especially in patients with concurrent kidney impairment (Ref).

### **Liver impairment prior to treatment initiation:**

#### ***Initial or dose adjustment in patients with preexisting liver cirrhosis***

**Child-Turcotte-Pugh class A: Oral, Parenteral, Rectal:** No dosage adjustment necessary; however, extend the dosing interval to a clinically appropriate frequency based on the indication and route of administration (eg, if the usual dosing interval is every 2 to 4 hours, extending the interval to every 4 to 6 hours may be appropriate) (Ref).

**Child-Turcotte-Pugh class B: Oral, Parenteral, Rectal:** Reduce dose to 50% of the usual indication-specific dose and extend the dosing interval to a clinically appropriate frequency based on the indication and route of administration (eg, if the usual dose and frequency is 2 mg IV every 2 to 4 hours, reducing the dose to 1 mg IV and extending the interval to every 4 to 6 hours may be appropriate) (Ref).

**Child-Turcotte-Pugh class C: Oral, Parenteral, Rectal:** Use not recommended; consider alternative agents that do not have active metabolites (Ref). If use of morphine is deemed necessary, reduce dose to  $\geq 50\%$  of the usual indication specific dose and extend the dosing interval to a clinically appropriate frequency based on the indication and route of administration (eg, if the usual dose and frequency is 2 mg IV every 2 to 4 hours, reducing the dose to 1 mg IV and extending the interval to every 6 to 8 hours may be appropriate) (Ref).

#### **Liver impairment developing in patients *already receiving morphine***

##### ***Chronic disease progression (eg, outpatient):***

**Baseline to Child-Turcotte-Pugh class A to C: Oral, Parenteral, Rectal:** No dosage adjustment necessary. Monitor closely for oversedation and/or new onset or worsening encephalopathy; if present, consider switching to an alternative agent. If use of morphine is deemed necessary, reduce dose and extend dosing interval to  $\geq 50\%$  of the usual indication-specific dose and frequency (eg, if the usual dose and frequency is 2 mg IV every 3 to 4 hours, reducing the dose to 1 mg IV and extending the interval to every 6 to 8 hours may be appropriate) (Ref).

##### ***Acute worsening of liver function (eg, requiring hospitalization)***

**Baseline to Child-Turcotte-Pugh class A to C: Oral, Parenteral, Rectal:** Avoid use; convert to an alternative agent that does not have active metabolites (Ref).

Dosing: Older Adult

**Note:** Minimize opioid use in older adults unless for the management of severe acute pain. Opioids are associated with an increased risk of falls and inducing or worsening delirium in older adults (Ref).

Refer to adult dosing. Use with caution; may require reduced dosage.

Dosing: Pediatric

(For additional information see "Morphine: Pediatric drug information")

#### **Dosage guidance:**

**Safety: Doses should be titrated to appropriate effect;** use lowest effective dose for the shortest duration to meet treatment goals. Assess need and discuss availability of naloxone for emergent treatment of opioid overdose with patient and/or caregiver upon initiating and renewing treatment. Dosage forms (parenteral and oral) are available in multiple concentrations; precautions should be taken to avoid confusion between the different concentrations; medication orders (prescriptions) should have the concentration specified as well as the dose clearly represented as milligram (mg) of morphine, not volume (mL). Use caution when changing routes of administration, potency differs between some products.

Acute pain, severe

**Acute pain, severe (opioid-naïve patients): Note:** Repeated SUBQ administration may cause local tissue irritation, pain, and induration. The use of IM injections is no longer recommended, especially for repeated administration due to painful administration, variable absorption, and lag time to peak effect; other routes are more reliable and less painful (Ref).

Infants  $\leq 6$  months:

**Note:** Infants  $< 3$  months are more susceptible to respiratory depression and cardiovascular adverse effects; use extra caution and consider frequent or continuous cardiorespiratory monitoring and administer in a setting that permits rapid management of respiratory insufficiency (Ref).

Intermittent dosing: Infants  $\leq 6$  months:

Oral: Oral solution (2 mg/mL or 4 mg/mL): Initial: 0.08 to 0.1 mg/kg/dose every 3 to 4 hours (Ref).

IV or SUBQ: Initial: 0.025 to 0.05 mg/kg/dose every 2 to 4 hours (Ref).

Continuous IV infusion: Infants  $\leq$  6 months: Initial: 0.008 to 0.02 mg/kg/**hour** (8 to 20 **mcg/kg/hour**); titrate carefully to effect (Ref). Higher initial doses of 0.05 mg/kg/**hour** (50 **mcg/kg/hour**) have been recommended in critically ill patients in the ICU (Ref).

Infants >6 months, Children, and Adolescents:

Intermittent dosing:

Oral: Infants >6 months, Children, and Adolescents:

Weight <50 kg: Oral: Immediate-release tablets, oral solution: Initial: 0.15 to 0.3 mg/kg/dose every 3 to 4 hours as needed; some experts have recommended higher initial doses of up to 0.5 mg/kg; usual initial maximum dose: 10 to 20 mg/dose (Ref).

Weight  $\geq$  50 kg: Oral: Immediate-release tablets, oral solution: 10 to 20 mg every 3 to 4 hours as needed (Ref).

IV, SUBQ: Infants >6 months, Children, and Adolescents:

Weight <50 kg: IV, SUBQ: Initial: 0.05 to 0.1 mg/kg/dose every 2 to 4 hours; titrate based on patient response (Ref).

Weight  $\geq$  50 kg: IV, SUBQ: Initial: 2 to 5 mg every 2 to 4 hours as needed. Use lower end of the dosing range in opioid naive patients; higher doses have been recommended (5 to 8 mg every 2 to 4 hours as needed) (Ref).

Continuous IV infusion, SUBQ continuous infusion: **Note:** Patients should have continuous respiratory monitoring (eg, pulse oximetry) and be in a setting that permits rapid management of respiratory insufficiency (Ref).

Infants >6 months, Children, and Adolescents:

Weight <50 kg: Initial: 0.01 mg/kg/**hour** (10 **mcg/kg/hour**); titrate carefully to effect; dosage range: 0.01 to 0.04 mg/kg/**hour** (10 to 40 **mcg/kg/hour**) (Ref). Higher initial doses of 0.05 mg/kg/**hour** (50 **mcg/kg/hour**) have been recommended in critically ill patients in the ICU (Ref).

Weight  $\geq$  50 kg: Initial: 1.5 mg/**hour**; titrate carefully based on clinical response; usual maintenance dose: 0.8 to 3 mg/**hour** (Ref). Higher initial doses of 2 mg/**hour** have been recommended in critically ill patients in the ICU (Ref).

Conversion from intermittent IV morphine: Administer the patient's total daily IV morphine dose over 24 hours as a continuous infusion; titrate dose to appropriate effect.

*Tapering after prolonged therapy (ie, physical dependence):* **Note:** Do **not** discontinue morphine abruptly; optimal taper schedule not established; individualize taper based on patient response; use of a withdrawal assessment tool is recommended; manufacturer labeling recommends:

Continuous IV infusion: Infants, Children, and Adolescents: Decrease dose by 25% to 50% every 2 to 4 days; monitor closely for signs and symptoms of withdrawal with each reduction in dose.

Oral solution: Children  $\geq$  2 years and Adolescents: Decrease dose by no greater than 10% to 25% of total daily dose every 2 to 4 weeks; monitor closely for signs and symptoms of withdrawal with each reduction in dose. Patients who have been taking opioids for short periods of time may tolerate a more rapid taper.

Chronic pain

### **Chronic pain:**

**Note:** There is no optimal or maximal dose for morphine in chronic pain. The appropriate dose is one that relieves pain throughout its dosing interval without causing unmanageable side effects. Consider total daily dose, potency, prior opioid use, degree of opioid experience and tolerance, conversion from previous opioid (including opioid formulation), patient's general condition, concurrent medications,

and type and severity of pain during prescribing process. Assess need and discuss availability of naloxone for emergent treatment of opioid overdose with patient and/or caregiver upon initiating and renewing treatment.

Oral: **Note:** A patient's morphine requirement should be established using immediate-release formulations (Ref). Conversion to long-acting products may be considered when chronic, continuous treatment is required. Higher dosages should be reserved for use only in opioid-tolerant patients.

*Tablets, extended release (eg, MS Contin):* Limited data available: Children and Adolescents able to swallow tablets whole:

Fixed dosing (Ref):

Weight 20 to <35 kg: 10 to 15 mg every 8 to 12 hours; **Note:** 10 mg strength not available in the United States.

Weight 35 to <50 kg: 15 to 30 mg every 8 to 12 hours.

Weight  $\geq$ 50 kg: 30 to 45 mg every 8 to 12 hours.

Weight-directed dosing: 0.3 to 0.6 mg/kg/dose every 12 hours (Ref).

*Discontinuation of extended-release formulations:* In general, gradually titrate dose downward (eg, decrease dose 10% to 25% every 2 to 4 weeks). Do not discontinue abruptly.

*Conversion from parenteral morphine or other opioids to extended-release formulation:* Substantial interpatient variability exists in relative potency. Therefore, it is safer to underestimate a patient's daily oral morphine requirement rather than overestimate requirements. Consider the parenteral to oral morphine ratio or other parenteral opioids to oral morphine conversions; divide total daily oral morphine dose into the appropriate interval for desired extended-release product.

*Conversion from other oral morphine formulations to extended-release formulation:* Children and Adolescents who can swallow whole tablets: Total daily oral morphine dose may be administered either in 2 divided doses daily (every 12 hours) **or** in 3 divided doses (every 8 hours) (Ref).

Continuous IV infusion: Children and Adolescents: Initial: 0.01 to 0.04 mg/kg/hour (10 to 40 mcg/kg/hour) (Ref); opioid-tolerant patients may require higher doses (Ref).

Epidural

**Epidural :** Limited data available: **Note: Must use preservative-free formulation:**

Intermittent: Infants, Children, and Adolescents: Epidural: 0.015 to 0.05 mg/kg/dose (15 to 50 mcg/kg/dose) (Ref); epidural administration can provide pain relief for 12 to 24 hours (Ref). Maximum dose: 0.1 mg/kg (100 mcg/kg) or 5 mg per 24 hours (Ref).

Continuous epidural infusion: Infants >6 months, Children, and Adolescents: Epidural: 0.001 to 0.005 mg/kg/hour (1 to 5 mcg/kg/hour) (Ref).

Palliative care, dyspnea management

**Palliative care, dyspnea management:**

*Inhalation (nebulization; preservative-free injection):* **Note:** Dose should be individualized and is dependent upon patient's previous or current systemic opioid exposure; doses not intended to provide analgesic activity; current systemic analgesia should be continued:

Infants, Children, and Adolescents: Very limited data available, efficacy results variable:

Initial dose: Inhalation: Equivalent to patient's 4-hour systemic morphine requirement (eg, IV or oral dose); titrate to effect (Ref); every 4 to 6 hour administration has been suggested (Ref). In the only pediatric case report (end-stage cystic fibrosis, age: 10 years, weight: 20 kg), an initial dose of 2.5 mg was used and the final dose was 10 mg every 4 to 6 hours (Ref).

Oral:

Infants, Children, and Adolescents: Limited data available:

Immediate-release preparations: Oral: Initial: 0.075 to 0.15 mg/kg/dose every 4 hours as needed (Ref); titrate for comfort.

*Continuous IV or SUBQ infusion:* **Note:** Use is recommended when oral therapy is ineffective:

Infants, Children, and Adolescents: Limited data available:

Continuous IV or SUBQ infusion: Initial: 0.005 mg/kg/**hour** (5 **mcg**/kg/**hour**); titrate for comfort (Ref); dosing based on palliative management of terminal infants with spinal muscular atrophy (type 1).

Patient-controlled analgesia, opioid naive

**Patient-controlled analgesia (PCA), opioid-naive:** **Note:** All patients should receive an initial loading dose of an analgesic (to attain adequate control of pain) before starting PCA for maintenance. Adjust doses, lockouts, and limits based on required loading dose, age, state of health, and presence of opioid tolerance. Assess patient and pain control at regular intervals and adjust settings if needed (Ref):

Children  $\geq 5$  years and Adolescents, weighing  $< 50$  kg: **Note:** PCA has been used in children as young as 5 years of age; however, clinicians need to assess children 5 to 8 years of age to determine if they are able to use the PCA device correctly (Ref).

Usual IV concentration: 1 mg/mL.

Demand dose: Usual initial: IV: 0.02 mg/kg/dose; usual range: 0.01 to 0.03 mg/kg/dose.

Lockout: Usual initial: 5 doses/**hour**.

Lockout interval: Range: 6 to 8 minutes.

Usual basal rate: IV: 0 to 0.03 mg/kg/**hour**.

Children  $\geq 5$  years and Adolescents, weighing  $\geq 50$  kg: **Note:** PCA has been used in children as young as 5 years of age; however, clinicians need to assess children 5 to 8 years of age to determine if they are able to use the PCA device correctly (Ref).

Usual IV concentration: 1 mg/mL.

Demand dose: Usual initial: IV: 1 mg; usual range: 0.5 to 2.5 mg.

Lockout interval: Usual initial: 6 minutes; usual range: 5 to 10 minutes.

Procedural pain and sedation

**Procedural pain and sedation:** Infants, Children, and Adolescents: IV: 0.05 to 0.1 mg/kg/dose; consider lowest dose in older children and adolescents; administer 5 minutes before the procedure; maximum dose: 4 mg/dose; may repeat dose in 5 minutes if necessary (Ref).

Tetralogy of fallot, hypercyanotic spell

**Tetralogy of fallot, hypercyanotic spell (infundibular spasm):** Limited data available:

Infants and Children:

IV: 0.1 mg/kg as a single dose (Ref).

IM: 0.1 to 0.2 mg/kg as a single dose (Ref).

SUBQ: 0.2 mg/kg as a single dose (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children and Adolescents: There are no specific dosage adjustments provided in the manufacturer's labeling. In general, the manufacturers recommend starting cautiously with lower doses, titrating slowly while carefully monitoring for side effects. However, the choice of an alternate opioid

may be prudent in patients with baseline renal impairment or rapidly changing renal function especially since other analgesics may be safer and reduced initial morphine dosing may result in suboptimal analgesia. Although clearance of morphine is similar to patients with normal renal function, morphine glucuronide metabolites (M3G [inactive as an analgesic; may contribute to CNS stimulation] and M6G [active analgesic]) accumulate in renal impairment resulting in increased sensitivity; patients may experience severe and prolonged respiratory depression which may even be delayed (Ref).

#### Dosing: Liver Impairment: Pediatric

Infants, Children and Adolescents: There are no dosage adjustments provided in manufacturer's labeling. Pharmacokinetics are unchanged in mild liver disease; substantial extrahepatic metabolism may occur. In patients with cirrhosis, the manufacturers recommend starting cautiously with lower doses, titrating slowly while carefully monitoring for adverse effects.

#### Adverse Reactions (Significant): Considerations

##### Opioid-induced constipation

Opioid-induced **constipation** (OIC) is the most common subtype of opioid-induced bowel dysfunction (OIBD), which is a broader term that encompasses additional GI opioid-induced adverse reactions, including nausea, vomiting, and gastroesophageal reflux. Symptoms of OIC may include decreased frequency of bowel movements, straining to pass bowel movements, a sense of incomplete evacuation, and/or hard stools (Ref). Tolerance does not develop to OIC (Ref). Symptoms are reversible after discontinuation of the opioid (Ref). OIC is the most common reason for discontinuation reported by patients and may often result in a longer hospital stay and increased overall healthcare costs (Ref). Opioid dose or route of administration do not appear to alter risk (Ref).

*Mechanism:* Time-related; mu-opioid receptor stimulation in the GI tract results in delayed gastric emptying, decreased peristalsis, decreased water and chloride secretion into the intestinal lumen, and slowed bowel motility (Ref).

*Onset:* Varied; changes in peristalsis may occur 5 to 25 minutes after administration of opioids (Ref). However, OIC is defined based on a 7-day period of change (Ref).

##### *Risk factors:*

- Increased dosing frequency (ie, daily vs 2 to 3 times per week) (Ref)
- Chronic opioid administration (Ref)
- Concurrent use of other medications that cause constipation

##### Opioid-induced neurotoxicity

Opioid-induced neurotoxicity is an opioid-related adverse drug reaction more commonly associated with opioids that have active metabolites (eg, codeine, hydromorphone, meperidine, morphine, oxycodone). Symptoms may include allodynia, delirium, hallucinations, hyperalgesia, hypersomnolence, myoclonus, tremor, and seizures (Ref).

*Mechanism:* Not clearly established; hypothesized to result from an accumulation of neuroexcitatory opioid metabolites (Ref).

##### *Risk factors:*

- High opioid dosage (Ref)
- Dehydration (Ref)
- Kidney failure (Ref)
- Infection (Ref)
- Older age (Ref)

##### Opioid-induced pruritus

Reversible peripherally- and/or centrally-mediated opioid-induced **pruritus** (OIP) may occur with systemic or neuraxial (ie, epidural or intrathecal) opioid administration though, it is most common with neuraxial delivery (Ref).

*Mechanism:* Dose-related; central OIP (neuraxial administration) is a result of stimulation of mu-opioid receptors in the spinal cord and brain (Ref). Peripheral OIP (systemic administration) is a result of opioid-induced nonimmunologic histamine release from mast cells (Ref).

*Onset:* Rapid; OIP occurred within 1 to 3 hours of neuraxial administration (Ref).

*Risk factors:*

- Higher doses (Ref)
- Route of administration: Neuraxial > IV > oral (Ref)
- Peripartum neuraxial administration (Ref)
- Neuraxial administration in major orthopedic surgery (Ref)
- Use of extended-release formulations (Ref)

Opioid-induced respiratory depression

Serious, life-threatening, or fatal opioid-induced **respiratory depression** (OIRD) may occur with use of morphine in adult and pediatric patients. Effects include hypoventilation, hypoxia, hypercapnia and respiratory acidosis, as well as reduced ventilatory responses to hypoxia and hypercapnia (Ref).

*Mechanism:* Dose-related; stimulation of mu-opioid receptors in the brainstem leads to suppression of the respiratory control network and depression of normal hypoxic and hypercapnic ventilatory responses (Ref).

*Onset:* Rapid; OIRD reported from 5 minutes to 1.2 hours (Ref).

*Risk factors:*

- Higher doses (Ref)
- Higher doses of intraoperative opioids (ie, morphine-equivalent daily dose ~40 mg in opioid-naive patients) (Ref)
- Initiation of therapy or dose increase
- Rapid IV injection (ie, over <2 to 3 minutes)
- Opioid naive (Ref)
- Opioid misuse (Ref)
- Acute overdose
- Concurrent administration of benzodiazepines, alcohol, or other CNS depressants
- Postoperative patients with comorbid cardiac or respiratory disease (Ref)
- Age >60 years (Ref)
- Males (Ref)
- Sleep disorder (eg, obstructive sleep apnea) (Ref)
- Chronic pulmonary disease
- Cachexia
- Debilitation

Opioid-induced withdrawal

Abruptly stopping or reducing opioid use in patients with physical dependence on opioids can precipitate opioid-induced withdrawal (OIW) in adult and pediatric patients (Ref). Physical symptoms may

include nausea, vomiting, diarrhea, abdominal cramps, tachycardia, chills, muscle aches, bone pain, agitation, anxiety, and insomnia. The presence and severity of withdrawal symptoms are associated with an increased risk of nonfatal overdose and receptive syringe sharing in people who inject drugs (Ref). Avoidance of withdrawal symptoms may drive continued opioid use (Ref). Symptoms typically dissipate over 4 to 7 days (Ref).

*Mechanism:* Withdrawal; opioids bind to mu-opioid receptors on neurons in the locus coeruleus (LC) of the brainstem, causing decreased norepinephrine (NE) release. Upon abrupt discontinuation, the absence of opioid stimulation causes LC hyperactivity, excessive NE release, and subsequent autonomic hyperactivity (Ref).

*Onset:* Rapid; symptoms typically occur within 12 hours of discontinuation and peak at 36 to 72 hours; varies based on half-life of opioid (Ref).

*Risk factors:*

- Higher cumulative dose (Ref)
- Prolonged exposure to opioids (Ref)
- Initiation of mixed agonist/antagonist or partial agonist analgesics (eg, buprenorphine, butorphanol, nalbuphine, pentazocine) (Ref) in patients currently or recently taking full mu-opioid agonists may cause precipitated opioid withdrawal. (Precipitated withdrawal is similar to OIW but with a faster onset.) (Ref)
- Opioid use disorder (Ref)
- Abrupt discontinuation or dose reduction (Ref)
- Concurrent use of benzodiazepines (Ref) or antipsychotics (Ref)
- Age <6 years, especially infants <6 months of age (Ref)
- Preexisting cognitive impairment (Ref)
- Critical illness involving the CNS (Ref)

#### Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

1% to 10%:

Cardiovascular: Atrial fibrillation, bradycardia, chest pain (2%), edema, facial flushing, flushing, hypertension, hypotension, palpitations, peripheral edema (3%), syncope, tachycardia, vasodilation

Dermatologic: Diaphoresis (2%), pallor, pressure ulcer, pruritus, skin rash (3%)

Endocrine & metabolic: Amenorrhea, decreased libido, gynecomastia, hyponatremia, SIADH

Gastrointestinal: Abdominal pain (3%), anorexia (3%), biliary colic, constipation (9%), delayed gastric emptying, diarrhea (3%), dyspepsia, dysphagia, gastric atony, gastroesophageal reflux disease, hiccups, nausea (7%), vomiting (2%), xerostomia (3%)

Genitourinary: Erectile dysfunction, prolonged labor, urinary hesitancy, urinary retention, urine abnormality

Hematologic & oncologic: Anemia (2%), leukopenia (2%), thrombocytopenia

Nervous system: Abnormal dreams, agitation, amnesia, anxiety (5%), apathy, asthenia (2%), ataxia, changes in thinking, chills, confusion (4%), decreased cough reflex, depression, dizziness (6%), drowsiness (9%), euphoria, hallucination, headache (2%), hypoesthesia, insomnia, lack of concentration, lethargy, malaise, myoclonus, paresthesia, seizure, slurred speech, tremor (2%), vertigo, voice disorder, withdrawal syndrome

Neuromuscular & skeletal: Arthralgia, back pain, foot-drop, ostealgia

Ophthalmic: Amblyopia, blurred vision, conjunctivitis, diplopia, miosis, nystagmus disorder

Respiratory: Asthma, atelectasis, dyspnea (3%), flu-like symptoms, hypoventilation, hypoxia, pulmonary edema (includes noncardiogenic), respiratory depression (including severe respiratory depression), respiratory insufficiency, rhinitis

Miscellaneous: Accidental injury (2%), fever (2%)

Frequency not defined:

Cardiovascular: Circulatory depression, orthostatic hypotension, peripheral vascular insufficiency (IV), phlebitis (IV), presyncope, shock

Dermatologic: Hemorrhagic urticaria, skin irritation, xeroderma

Endocrine & metabolic: Antidiuretic effect, increased thirst, weight loss

Gastrointestinal: Decreased appetite, dysgeusia, gastroenteritis, gastrointestinal hypermotility (IV; in patients with chronic ulcerative colitis), rectal disease, toxic megacolon (IV; patients with chronic ulcerative colitis)

Genitourinary: Bladder spasm (IV), dysuria, ejaculatory disorder, hypogonadism, oliguria, ureteral spasm (IV)

Hypersensitivity: Nonimmune anaphylaxis

Local: Erythema at injection site (IV), induration at injection site (SUBQ), local swelling (IV, intrathecal, epidural; genital swelling in males following infusion device implant surgery), pain at injection site (SUBQ)

Nervous system: Abnormal gait, altered mental status, coma, delirium, disorientation, disruption of body temperature regulation (IV, epidural, intrathecal), drug abuse, dysphoria, fear, increased intracranial pressure, mood changes, neonatal withdrawal, nervousness, paradoxical central nervous system stimulation (IV, epidural, intrathecal), sedated state, toxic psychosis (IM, IV, epidural, intrathecal)

Neuromuscular & skeletal: Decreased bone mineral density, laryngospasm, muscle rigidity, muscle spasm (IV, epidural, intrathecal; myoclonic spasm of lower extremities), muscle twitching

Ophthalmic: Eye pain, visual disturbance

Respiratory: Apnea

Miscellaneous: Granuloma (inflammatory mass: IV, epidural, intrathecal), impaired physical performance

Postmarketing (any route of administration):

Dermatologic: Urticaria (Ref)

Endocrine & metabolic: Hyperprolactinemia (Ref)

Gastrointestinal: Intestinal obstruction

Hepatic: Increased liver enzymes

Hypersensitivity: Anaphylaxis (Ref)

Nervous system: Allodynia (opioid-induced hyperalgesia) (Ref), difficulty thinking, fatigue, hyperalgesia, hypertonia, hypothermia (Ref), opioid dependence (Ref)

Respiratory: Bronchospasm

Contraindications

Hypersensitivity (eg, anaphylaxis) to morphine or any component of the formulation; significant respiratory depression; acute or severe bronchial asthma in an unmonitored setting or in the absence of resuscitative equipment; concurrent use of monoamine oxidase inhibitors (MAOIs) or use of MAOIs within the last 14 days; GI obstruction, including paralytic ileus (known or suspected).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Additional contraindications:

Epidural/intrathecal: Infection at infusion site; concomitant anticoagulant therapy; uncontrolled bleeding diathesis; presence of any other concomitant therapy or medical condition that would render administration hazardous; upper airway obstruction.

*Canadian labeling:* Additional contraindications (not in US labeling):

Contraindications may vary per product labeling; refer also to product labels: Suspected surgical abdomen (eg, acute appendicitis, pancreatitis); disease/condition that affects bowel transit (known or suspected); chronic obstructive airway; status asthmaticus; management of acute pain, including use in outpatient or day surgeries; mild, intermittent, or short-duration pain that can be managed with other pain medications; acute respiratory depression; hypercarbia; cor pulmonale; acute alcoholism; delirium tremens; convulsive disorders; severe CNS depression; increased cerebrospinal; intracranial pressure; brain tumor; head injury; cardiac arrhythmias; pregnancy; use during labor and delivery; breastfeeding; alcohol consumption or medications containing alcohol (Kadian, M-Eslon product monographs); toxic psychosis and severe kyphoscoliosis (Kadian product monograph); severe cirrhosis (M-Ediat product monograph); hypotension (M-Eslon product monograph), emotional instability and/or suicidal ideation (Statex product monograph); surgical anastomosis (morphine sulfate inj Sandoz product monograph)

Warnings/Precautions

**Concerns related to adverse effects:**

- **CNS depression:** May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery or driving). Carbon dioxide retention from opioid-induced respiratory depression can exacerbate the sedating effects of opioids. Some dosage forms may be contraindicated in patients with severe CNS depression.
- **Esophageal dysfunction:** Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.
- **Hyperalgesia:** Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.
- **Hypotension:** May cause severe hypotension (including orthostatic hypotension and syncope); use with caution in patients with hypovolemia, cardiovascular disease (including acute MI), circulatory shock, or drugs that may exaggerate hypotensive effects (including phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration. Avoid use in patients with circulatory shock. Some dosage forms may be contraindicated in patients with cardiac arrhythmias or heart failure due to chronic lung disease.
- **Phenanthrene hypersensitivity:** Use with caution in patients with hypersensitivity reactions to other phenanthrene derivative opioid agonists (codeine, hydrocodone, hydromorphone, levorphanol, oxycodone, oxymorphone).

**Disease-related concerns:**

- **Abdominal conditions:** May obscure diagnosis or clinical course of patients with acute abdominal conditions.
- **Adrenocortical insufficiency:** Use with caution in patients with adrenal insufficiency, including Addison disease. Long-term opioid use may cause secondary hypogonadism, which may lead to sexual dysfunction, infertility, mood disorders, and osteoporosis (Brennan 2013).

- Biliary tract impairment: Use with caution in patients with biliary tract dysfunction or acute pancreatitis; opioids may cause constriction of sphincter of Oddi.
- Cardiovascular conditions: Morphine may cause constipation, which may be problematic in patients with unstable angina and patients post-myocardial infarction.
- CNS depression/coma: Avoid use in patients with impaired consciousness or coma, as these patients are susceptible to intracranial effects of CO<sub>2</sub> retention.
- Delirium tremens: Use with caution in patients with delirium tremens. Some dosage forms may be contraindicated in patients with delirium tremens.
- Head trauma: Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur. Some dosage forms may be contraindicated in patients with increased intracranial or cerebrospinal pressure, head injuries, or brain tumor.
- Hepatic impairment: Use with caution in patients with severe hepatic impairment.
- Mental health conditions: Use opioids with caution for chronic pain in patients with mental health conditions (eg, depression, anxiety disorders, post-traumatic stress disorder) due to potential increased risk for opioid use disorder and overdose; more frequent monitoring is recommended (CDC [Dowell 2022]).
- Obesity: Use with caution in patients who are morbidly obese.
- Prostatic hyperplasia/urinary stricture: Use with caution in patients with prostatic hyperplasia and/or urinary stricture.
- Psychosis: Use with caution in patients with toxic psychosis.
- Renal impairment: Use with caution in patients with renal impairment.
- Respiratory disease: Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale and patients having a substantially decreased respiratory reserve, hypoxia, hypercarbia, or preexisting respiratory depression, particularly when initiating therapy and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages. Consider the use of alternative nonopioid analgesics in these patients.
- Sleep-related disorders: Use with caution in patients with sleep-related disorders, including sleep apnea, due to increased risk for respiratory and CNS depression. Monitor carefully and titrate dosage cautiously in patients with mild sleep-disordered breathing. Avoid opioids in patients with moderate to severe sleep-disordered breathing (CDC [Dowell 2022]).
- Seizure disorders: Use with caution in patients with seizure disorders; may cause or exacerbate preexisting seizures. Some dosage forms may be contraindicated in patients with seizure disorder.
- Thyroid dysfunction: Use with caution in patients with thyroid dysfunction.

#### ***Concurrent drug therapy issues:***

- Benzodiazepines or other CNS depressants: Some dosage forms may be contraindicated in patients with acute alcoholism.

#### ***Special populations:***

- Older adult: Use opioids with caution in older adults; may be more sensitive to adverse effects. Clearance may also be reduced in older adults (with or without renal impairment), resulting in a narrow therapeutic window and increased adverse effects. Monitor closely for adverse effects associated with opioid therapy (eg, respiratory and CNS depression, falls, cognitive impairment, constipation) (CDC [Dowell 2022]). Consider the use of alternative nonopioid analgesics in these patients when possible.

#### ***Dosage form specific issues:***

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol ( $\geq 99$  mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [“Inactive” 1997]; CDC, 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.
- ER formulations: Therapy should only be prescribed by health care professionals familiar with the use of potent opioids for chronic pain. ER products are not interchangeable. When determining a generic equivalent or switching from one ER product to another, a thorough understanding of the pharmacokinetic properties is important in determining the proper generic equivalent or proper dose of the other ER product (review of the manufacturer’s label may be necessary).
- Infumorph, Duramorph, Mitigo: Neuroaxial administration: Opioid overdose reversal agents (eg, naloxone, nalmeferene) should be immediately available. Thoracic epidural administration has been shown to dramatically increase the risk of early and late respiratory depression. High doses ( $>20$  mg/day) of neuraxial morphine may produce myoclonic events. Patients with reduced circulating blood volume or impaired myocardial function, or on concomitant sympatholytic drugs should be monitored for orthostatic hypotension, a frequent complication in single-dose neuraxial morphine.
- Infumorph, Mitigo: Should only be used in microinfusion devices; not for IV, IM, or SUBQ administration or for single-dose administration. Administer intrathecal doses of 10 and 25 mg/mL to the lumbar area. Monitor closely, especially in the first 24 hours. Inflammatory masses (eg, granulomas), some resulting in severe neurologic impairment, have occurred when receiving Infumorph or Mitigo via indwelling intrathecal catheter; monitor carefully for new neurologic signs/symptoms.
- Injections: Products are designed for administration by specific routes (ie, IV, intrathecal, epidural). Use caution when prescribing, dispensing, or administering to use formulations only by intended route(s). Rapid IV administration may result in chest wall rigidity. Use with caution when injecting IM into chilled areas or in patients with hypotension or shock (impaired perfusion may prevent complete absorption); if repeated injections are administered, an excessive amount may be suddenly absorbed if normal circulation is re-established.
- Product interchange: Improper or erroneous substitution of Infumorph or Mitigo for regular Duramorph is likely to result in serious overdosage, leading to seizures, respiratory depression, and possibly a fatal outcome.
- Sulfites: Some dosage forms may contain sulfites that may cause allergic reactions in sulfite sensitive patients.

#### **Other warnings/precautions:**

- Abuse/misuse/diversion: Use with caution in patients with a history of substance use disorder; potential for drug dependency exists. Other factors associated with increased risk for misuse include concomitant depression or other mental health conditions, higher opioid dosages, long-term use, or taking other CNS depressants. Consider offering an opioid overdose reversal agent (eg, naloxone, nalmeferene) prescription in patients with an increased risk for overdose, such as history of overdose or substance use disorder, higher opioid dosages ( $\geq 50$  morphine milligram equivalents [MME]/day orally), concomitant benzodiazepine use, and patients at risk for returning to a high dose after losing tolerance (CDC [Dowell 2022]).
- Appropriate use: Outpatient setting: Opioids should **not** be used as first-line therapy for acute ( $<1$  month duration), subacute (1 to 3 month duration), or chronic pain ( $>3$  month duration [outside of end-of-life or palliative care, active cancer treatment, sickle cell disease, or medication-based opioid use disorder treatment]). Preferred management includes nonpharmacologic therapy and nonopioid therapy (eg, nonsteroidal anti-inflammatory drugs, acetaminophen, certain antiseizure medications and antidepressants) as appropriate for the specific condition. If opioid therapy is initiated, it should be combined with nonpharmacologic and nonopioid therapy, as appropriate. Prior to initiation, known risks and realistic benefits of opioid therapy should be discussed with the patient. Therapy should

be initiated at the lowest effective dosage using IR opioids (instead of ER/long-acting opioids). For the treatment of acute pain, therapy should only be given for the expected duration of pain severe enough to require opioids and prescribed as needed (not scheduled). For the treatment of subacute and chronic pain, realistic treatment goals for pain/function should be established, including consideration for discontinuation if benefits do not outweigh risks. Therapy should be continued only if clinically meaningful improvement in pain/function outweighs risks. Risk to patients increases with higher opioid dosages. Dosages  $\geq 50$  MME/day are likely to not have increased benefit to pain relief or function relative to overall risk to patients; before increasing dosage to  $\geq 50$  MME/day, readdress pain and reassess evidence of individual benefits and risks (CDC [Dowell 2022]).

- Opioid overdose reversal agent: Discuss the availability of opioid overdose reversal agents (eg, naloxone, nalmefene) with all patients who are prescribed opioid analgesics, as well as their caregivers, and consider prescribing it to patients who are at increased risk of opioid overdose. These include patients who are also taking benzodiazepines or other CNS depressants, have an opioid use disorder (OUD) (current or history of) or substance use disorder, have higher opioid dosages ( $\geq 50$  MME/day orally) (CDC [Dowell 2022]), or have experienced opioid-induced respiratory depression/opioid overdose. Additionally, health care providers should consider prescribing an opioid overdose reversal agent to patients prescribed medications to treat OUD; patients at risk of opioid overdose even if they are not taking an opioid analgesic or medication to treat OUD; and patients taking opioids, including methadone or buprenorphine for OUD, if they have household members, including children, or other close contacts at risk for accidental ingestion or opioid overdose. Inform patients and caregivers on options for obtaining an opioid overdose reversal agent (eg, by prescription, OTC, directly from a pharmacist, a community-based program) as permitted by state dispensing and prescribing guidelines. Educate patients and caregivers on how to recognize respiratory depression, proper administration and important differences of opioid overdose reversal agents, and getting emergency help even if an opioid overdose reversal agent is administered.

- Optimal regimen: An opioid-containing analgesic regimen should be tailored to each patient's needs and based upon the type of pain being treated (acute versus chronic), the route of administration, degree of tolerance for opioids (naïve versus chronic user), age, weight, and medical condition. The optimal analgesic dose varies widely among patients; doses should be titrated to pain relief/prevention.

- Surgery: Opioids decrease bowel motility; monitor for decreased bowel motility in postoperative patients receiving opioids. Use with caution in the perioperative setting; individualize treatment when transitioning from parenteral to oral analgesics. Some dosage forms may be contraindicated after biliary tract surgery, suspected surgical abdomen, or surgical anastomosis.

#### Warnings: Additional Pediatric Considerations

Prolonged use of any morphine product during pregnancy can result in neonatal opioid withdrawal syndrome, which may be life-threatening if not recognized and treated, and requires management according to protocols developed by neonatology experts.

It is well known that recurrent exposure to painful stimuli without effective pain management alters the pain pathways in the neonatal brain and results in altered pain perception. There is a growing concern regarding the need to balance pain management with the concern for neuronal apoptosis and potential poor neurodevelopmental outcomes in neonatal patients treated with opiates. In vitro and animal studies show that exposure to morphine induces apoptosis of primary human fetal microglial cells (Attarian 2014; Hu 2002). Clinical evaluations of preterm neonates exposed to morphine have shown impaired cerebellar and cerebral growth but have shown mixed impact on the long-term neurodevelopment outcomes in early childhood (Steinhorn 2015; Zwicker 2016).

#### Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule Extended Release 24 Hour, Oral, as sulfate:

Generic: 10 mg [DSC], 20 mg [DSC], 30 mg, 45 mg, 50 mg [DSC], 60 mg, 75 mg, 80 mg [DSC], 90

mg, 100 mg [DSC], 120 mg

Solution, Injection, as sulfate [preservative free]:

Duramorph: 0.5 mg/mL (10 mL); 1 mg/mL (10 mL) [antioxidant free]

Infumorph 200: 200 mg/20 mL (10 mg/mL) (20 mL) [antioxidant free]

Infumorph 500: 500 mg/20 mL (25 mg/mL) (20 mL) [antioxidant free]

Mitigo: 200 mg/20 mL (10 mg/mL) (20 mL); 500 mg/20 mL (25 mg/mL) (20 mL)

Generic: 0.5 mg/mL (10 mL); 1 mg/mL (10 mL); 2 mg/mL (1 mL); 4 mg/mL (1 mL); 5 mg/mL (1 mL [DSC]); 8 mg/mL (1 mL [DSC]); 10 mg/mL (1 mL)

Solution, Intravenous, as sulfate:

Generic: 50 mg/mL (20 mL)

Solution, Intravenous, as sulfate [preservative free]:

Generic: 1 mg/mL (30 mL); 2 mg/mL (1 mL); 4 mg/mL (1 mL); 8 mg/mL (1 mL); 10 mg/mL (1 mL); 50 mg/mL (20 mL [DSC], 50 mL)

Solution, Oral, as sulfate:

Generic: 10 mg/5 mL (5 mL, 100 mL, 500 mL); 20 mg/5 mL (100 mL, 500 mL); 20 mg/mL (15 mL [DSC], 30 mL [DSC], 118 mL, 120 mL [DSC], 240 mL); 100 mg/5 mL (15 mL, 30 mL, 120 mL, 240 mL)

Suppository, Rectal, as sulfate:

Generic: 5 mg (12 ea); 10 mg (12 ea); 20 mg (12 ea); 30 mg (12 ea)

Tablet, Oral, as sulfate:

Generic: 15 mg, 30 mg

Tablet Extended Release, Oral, as sulfate:

MS Contin: 15 mg [contains fd&c blue #2 (indigotine, indigo carmine)]

MS Contin: 30 mg [contains fd&c blue #1 (brilliant blue)]

MS Contin: 60 mg [contains quinoline yellow (d&c yellow #10)]

MS Contin: 100 mg [DSC]

MS Contin: 200 mg [DSC] [contains fd&c blue #1 (brilliant blue), quinoline yellow (d&c yellow #10)]

Generic: 15 mg, 30 mg, 60 mg, 100 mg, 200 mg

Generic Equivalent Available: US

Yes

Pricing: US

**Capsule ER 24 Hour Therapy Pack** (Morphine Sulfate ER Beads Oral)

30 mg (per each): \$5.73

45 mg (per each): \$8.50

60 mg (per each): \$11.13

75 mg (per each): \$14.16

90 mg (per each): \$16.73

120 mg (per each): \$19.74

**Solution** (Infumorph 200 Injection)

200 MG/20ML (10 mg/mL) (per mL): \$12.48

**Solution** (Infumorph 500 Injection)

500 MG/20ML (25 mg/mL) (per mL): \$21.22

**Solution** (Mitigo Injection)

200 MG/20ML (10 mg/mL) (per mL): \$13.61

500 MG/20ML (25 mg/mL) (per mL): \$23.15

**Solution** (Morphine Sulfate (Concentrate) Oral)

100 mg/5 mL (per mL): \$0.35 - \$0.84

**Solution** (Morphine Sulfate (PF) Injection)

0.5 mg/mL (per mL): \$2.79

1 mg/mL (per mL): \$5.59

2 mg/mL (per mL): \$4.30

4 mg/mL (per mL): \$4.30

10 mg/mL (per mL): \$3.66

**Solution** (Morphine Sulfate (PF) Intravenous)

1 mg/mL (per mL): \$0.56

2 mg/mL (per mL): \$6.85 - \$12.00

4 mg/mL (per mL): \$10.80

10 mg/mL (per mL): \$33.60

**Solution** (Morphine Sulfate Injection)

2 mg/mL (per mL): \$4.30

4 mg/mL (per mL): \$4.30

**Solution** (Morphine Sulfate Intravenous)

2 mg/mL (per mL): \$3.10

4 mg/mL (per mL): \$2.10 - \$3.10

8 mg/mL (per mL): \$3.38 - \$3.39

10 mg/mL (per mL): \$3.06

50 mg/mL (per mL): \$5.22

**Solution** (Morphine Sulfate Oral)

10 mg/5 mL (per mL): \$0.44 - \$0.67

20 mg/5 mL (per mL): \$0.18 - \$0.19

**Suppository** (Morphine Sulfate Rectal)

5 mg (per each): \$5.00

10 mg (per each): \$6.25

20 mg (per each): \$7.50

30 mg (per each): \$9.38

**Tablet, controlled release** (Morphine Sulfate ER Oral)

15 mg (per each): \$0.56 - \$2.04

30 mg (per each): \$0.64 - \$3.68

60 mg (per each): \$1.05 - \$6.21

100 mg (per each): \$1.11 - \$9.20

200 mg (per each): \$2.73 - \$18.38

**Tablet, controlled release** (MS Contin Oral)

15 mg (per each): \$5.86

30 mg (per each): \$11.13

60 mg (per each): \$21.72

**Tablets** (Morphine Sulfate Oral)

15 mg (per each): \$0.52 - \$0.91

30 mg (per each): \$0.88 - \$1.54

**Disclaimer:** A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

**Dosage Forms: Canada**

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

**Capsule, Oral:**

M-Ediat: 5 mg [contains corn starch, fd&c blue #2 (indigotine, indigo carmine)]

M-Ediat: 10 mg [contains corn starch]

M-Ediat: 20 mg [contains corn starch, fd&c blue #1 (brilliant blue), fd&c red #40 (allura red ac dye)]

M-Ediat: 30 mg [contains corn starch, fd&c blue #2 (indigotine, indigo carmine), quinoline yellow (d&c yellow #10)]

**Capsule Extended Release 12 Hour, Oral:**

M-Eslon: 10 mg, 15 mg, 30 mg, 60 mg, 100 mg, 200 mg

**Capsule Extended Release 24 Hour, Oral, as sulfate:**

Kadian: 20 mg, 50 mg, 100 mg

Kadian SR: 10 mg

**Solution, Injection, as sulfate:**

Generic: 0.5 mg/mL (10 mL); 1 mg/mL (5 mL); 2 mg/mL (1 mL); 10 mg/mL (1 mL, 5 mL); 15 mg/mL (1 mL, 30 mL); 50 mg/mL (1 mL, 5 mL, 10 mL, 50 mL)

**Solution, Intravenous, as sulfate:**

Generic: 100 mg /100 mL in NaCl 0.9% (100 mL)

**Syrup, Oral:**

Doloral: 1 mg/mL (225 mL, 500 mL) [contains alcohol, usp, fd&c blue #1 (brilliant blue), fd&c yellow #5 (tartrazine), methylparaben (methyl p-hydroxybenzoate), propylparaben (propyl p-hydroxybenzoate)]

Doloral: 5 mg/mL (225 mL, 500 mL)

Tablet, Oral:

MS/IR: 5 mg, 10 mg, 20 mg

Statex: 5 mg, 10 mg [contains fd&c blue #1 (brill blue) aluminum lake]

Statex: 25 mg [DSC], 50 mg [DSC]

Generic: 5 mg, 10 mg

Tablet, Oral, as sulfate:

MS/IR: 30 mg

Tablet Extended Release, Oral, as sulfate:

MS Contin: 15 mg [contains fd&c blue #1 (brill blue) aluminum lake, fd&c blue #2 (indigotine, indigo carmine), quinoline (d&c yellow #10) aluminum lake]

MS Contin: 30 mg [contains fd&c blue #2 (indigo carm) aluminum lake, fd&c yellow #6 (sunset yellow) aluminum lake]

MS Contin: 60 mg [contains fd&c yellow #6 (sunset yellow) aluminum lake, quinoline (d&c yellow #10) aluminum lake]

MS Contin: 100 mg [contains fd&c blue #2 (indigo carm) aluminum lake]

MS Contin: 200 mg [contains fd&c yellow #6 (sunset yellow) aluminum lake]

Generic: 15 mg, 30 mg, 60 mg, 100 mg, 200 mg

Extemporaneous Preparations

#### **0.4 mg/mL (400 mcg/mL) Oral Solution**

A 0.4 mg/mL oral solution may be made using the **2 mg/mL** oral morphine solution. Measure 10 mL (20 mg) of the 2 mg/mL oral morphine solution and transfer to a plastic amber bottle. Measure 40 mL of sterile water for irrigation and add to bottle containing the morphine. Shake to mix. Store at room temperature. Stable for 60 days (Ref).

Controlled Substance

C-II

Administration: Adult

**Oral:** Do not crush, chew, or dissolve ER formulations; swallow whole. Cutting, breaking, crushing, chewing, or dissolving ER formulations may result in uncontrolled delivery of morphine, leading to overdose and death.

Product-specific administration information:

ER capsules (once-daily formulation [pellets]): Capsules may be opened and sprinkled on applesauce and eaten immediately without chewing; do not crush, dissolve, or chew the beads, as it can result in a rapid release of a potentially fatal dose of morphine. Ensure all pellets have been swallowed by rinsing mouth. Do not administer through gastric/nasogastric tubes.

ER capsules (once- or twice-daily formulation [polymer coated pellets]): Capsules may be opened and sprinkled on applesauce and eaten immediately without chewing; do not crush, dissolve, or chew the beads, as it can result in a rapid release of a potentially fatal dose of morphine. Ensure all pellets have been swallowed by rinsing mouth. Contents of capsules may be opened and sprinkled over 10 mL water and flushed through prewetted 16F gastrostomy tube; use swirling motion to pour pellets and water into the gastrostomy tube through a funnel. Rinse beaker with a further 10 mL of water

and pour into the funnel; repeat rinsing until no pellets remain in the beaker. Do not administer through nasogastric tubes.

**Bariatric surgery:** Tablet, extended release: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. ER tablet should be swallowed whole. Do not break, crush, cut, chew, dissolve, presoak, wet, or split as this will result in uncontrolled delivery of morphine that can lead to overdose or death. Take one abuse deterrent dose at a time and with sufficient water to avoid tablet becoming sticky and difficult to swallow. IR tablet, oral solution, and rectal suppository are available. If safety and efficacy can be effectively monitored, no change in formulation or administration is required after bariatric surgery; however, clinicians should be advised that oral morphine has been shown to have significantly increased  $C_{max}$  and decreased  $T_{max}$  in the immediate (1 to 2 weeks) and long-term (6 months) periods after bariatric surgery (Ref).

**IV:** Administer single-use prefilled syringes/cartridges via slow IV push over 4 to 5 minutes (rapid administration may result in chest wall rigidity). Concentrated vials are available for preparation of continuous IV infusion or PCA; refer to indication-specific infusion rates in dosing for detailed recommendations.

**Epidural, intrathecal:** Use only preservative-free solutions indicated for intrathecal or epidural use; refer to indication-specific infusion rates in dosing for detailed recommendations.

**Rectal:** Remove suppository from plastic packet and moisten suppository with water to avoid irritation. Gently insert (rounded end first) approximately a finger's length into rectum, angling it toward the umbilicus, and placing it against the rectal wall; after suppository is inserted, hold buttocks together until urge to expel ceases (Ref).

**SUBQ continuous infusions:** Change site every 3 to 7 days or when erythema occurs (Ref). To maintain comfort, the SUBQ infusion rate should generally not exceed 5 mL/hour (Ref).

Preparation for Administration: Adult

**Parenteral:** Solutions for injection should be visually inspected for particulate matter and discoloration prior to preparation. Do not use if solution contains a precipitate or is darker in color than pale yellow or discolored in any other way.

**IV push/intermittent infusion:** May dilute to a final concentration of 0.5 to 5 mg/mL

**Continuous IV infusion:** Dilute in D5W or NS to a usual final concentration of 0.1 to 5 mg/mL; more concentrated solutions may be used in patients requiring fluid restriction or high doses (Ref); concentrations >5 mg/mL are rarely needed (Ref).

**Epidural and intrathecal:** Use only preservative-free injections indicated for epidural/intrathecal use. Dilution may be required (preservative-free NS recommended); determined by the individual dosage requirements and the characteristics of the continuous microinfusion device; some products (eg, Infumorph) require filtering through ≤5 micron microfilter before injecting into microinfusion device.

Usual Infusion Concentrations: Adult

**IV infusion:** 1 mg/mL

Administration: Pediatric

Oral: Administer with food.

**Immediate release:** Oral solution: Available in multiple strengths, including a concentrated oral solution (20 mg/mL). Precautions should be taken to avoid confusion between the different concentrations; medication orders (prescriptions) should have the concentration specified as well as the dose clearly represented as milligram (mg) of morphine, not volume (mL). Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

**Concentrated oral solution:** Only use in opioid-tolerant patients (eg, adults taking ≥60 mg/day of morphine or equivalent for ≥1 week). Administer only using the calibrated oral syringe provided

with the concentrated oral solution to ensure the dose is measured and administered accurately.

Extended-/controlled-release products: Do not chew, crush, break, or dissolve extended- and controlled-release products as it can result in uncontrolled delivery and can lead to overdose or death; swallow whole.

Parenteral: **Note:** Solutions for injection should be visually inspected for particulate matter and discoloration prior to administration. Do not use if contains a precipitate or is darker in color than pale yellow or discolored in any other way. Precautions should be taken to avoid confusion between the different concentrations; medication orders (prescriptions) should have the concentration specified as well as the dose clearly represented as milligram (mg) of morphine, not volume (mL).

IV push: Administer undiluted or diluted solution over 4 to 5 minutes (Ref); rapid IV administration may increase adverse effects. For neonates, IV boluses may be administered over 5 to 15 minutes via a syringe pump (Ref).

Intermittent IV infusion: Further dilute and administer over 15 to 30 minutes.

Continuous IV infusion: Administer as a continuous infusion via an infusion pump.

Epidural and intrathecal: **Use only preservative-free injections.**

Inhalation: Limited data available; utilize preservative free parenteral morphine and further dilute prior to administration (Ref)

Preparation for Administration: Pediatric

Parenteral:

IV push/intermittent infusion: May dilute to a final concentration of 0.5 to 5 mg/mL. Vermont Oxford Network recommends standard concentrations of 0.1 mg/mL or 0.5 mg/mL for intermittent doses for neonates (Ref).

Continuous IV infusion: Dilute in D5W, D10W, or NS to a usual final concentration of 0.04 to 1 mg/mL; more concentrated solutions (eg, 5 mg/mL) may be used in patients requiring fluid restriction or high doses (Ref); concentrations >5 mg/mL are rarely needed (Ref). Vermont Oxford Network recommends a standard concentration of 0.1 mg/mL or 0.5 mg/mL for neonates (Ref); for pediatric patients, ASHP recommends standard concentrations of 0.2 mg/mL, 0.5 mg/mL, or 1 mg/mL (Ref).

Epidural and intrathecal: **Use only preservative-free injections.** Dilution may be required; determined by the individual patient's dosage requirements and the characteristics of the continuous microinfusion device; filter through  $\leq 5$  micron microfilter before injecting into microinfusion device.

Inhalation: Limited data available; utilize preservative-free parenteral morphine and dilute with NS to a final volume of 5 mL (Ref).

Usual Infusion Concentrations: Pediatric

**IV infusion:** 0.2 mg/mL, 0.1 mg/mL, 0.5 mg/mL, **or** 1 mg/mL.

Storage/Stability

Parenteral: Store intact vials at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Do not freeze. Store in carton until use. Protect from light. Degradation depends on pH and presence of oxygen; relatively stable in pH  $\leq 4$ . Darkening of solutions indicate degradation. Do not heat-sterilize. Once diluted may store NS preparations at room temperature 20°C to 25°C (68°F to 77°F) or in refrigerator 2°C to 8°C (36°F to 46°F) for up to 14 days; D5W preparations may be stored at room temperature 20°C to 25°C (68°F to 77°F) for up to 72 hours or in refrigerator 2°C to 8°C (36°F to 46°F) for up to 14 days.

Oral:

Extended release: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect capsules from light and moisture.

Immediate release: Store at 20°C to 25°C (68°F to 77°F). Protect from moisture.

Suppositories: Store at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

**Pain management , acute and chronic pain:**

Injection: Management of pain severe enough to require an opioid analgesic and for which alternative treatments are inadequate.

Preservative-free solutions only:

*Duramorph*: Epidural or intrathecal management of pain without attendant loss of motor, sensory, or sympathetic function. **Note:** Not for use in continuous microinfusion devices.

*Infumorph, Mitigo*: Used in continuous microinfusion devices for intrathecal or epidural administration in management of intractable chronic pain severe enough to require an opioid analgesic and for which less invasive means of controlling pain are inadequate. **Note:** Not for single-dose IV, IM, or subcutaneous administration or single-dose neuraxial injection.

Oral:

*Extended release*: Management of severe and persistent pain that requires an opioid analgesic and that cannot be adequately treated with alternative options, including IR opioids.

*Immediate release*:

Oral solution: Management of acute pain (pediatric patients  $\geq 2$  years of age and adults) and chronic pain (adults only) severe enough to require an opioid analgesic and for which alternative treatments are inadequate. Oral solution 100 mg per 5 mL (20 mg/mL) is for opioid-tolerant adults only.

Tablets: Management of acute pain (pediatric patients  $\geq 50$  kg and adults) and chronic pain (adults only) severe enough to require an opioid analgesic and for which alternative treatments are inadequate.

Rectal: Management of acute and chronic pain severe enough to require an opioid analgesic and for which alternative treatments are inadequate.

Limitations of use: Because of the risks of substance use disorder, abuse, misuse, overdose, and death, which may occur at any dosage or duration and persist over the course of therapy, reserve opioid analgesics, including morphine, for use in patients for whom alternative treatment options are ineffective, not tolerated, or would be otherwise inadequate to provide sufficient management of pain. ER formulations are not indicated as as-needed analgesics.

**Note:** "Opioid-tolerant" patients are defined as patients who are taking at least:

Oral morphine 60 mg/day, **or**

Transdermal fentanyl 25 mcg/hour, **or**

Oral oxycodone 30 mg/day, **or**

Oral hydromorphone 8 mg/day, **or**

Oral oxymorphone 25 mg/day, **or**

Oral hydrocodone 60 mg/day, **or**

Equianalgesic dose of another opioid for at least 1 week

Use: Off-Label: Adult

Cough in select patients; Critically ill patients in the ICU (analgesia and sedation); Dyspnea in palliative care patients

Medication Safety Issues

Sound-alike/look-alike issues:

Morphine may be confused with diamorphine, HYDROmorphine, methadone

Morphine sulfate may be confused with magnesium sulfate

MS Contin may be confused with OxyCONTIN

Roxanol may be confused with OxyFast, Roxicet, Roxicodone Intensol

Older Adult: High-Risk Medication:

Morphine is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults ( $\geq 65$  years of age) due to an increased risk of falls and exacerbation of constipation. Opioids are not recommended as first-line treatment of mild pain or long-term treatment of osteoarthritis. Short-acting opioids should be available for breakthrough pain in patients on long-acting opioids (O'Mahony 2023).

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (epidural and intrathecal medications; opioids, all formulations and routes of administration [with special emphasis on concentrated morphine solution 20 mg/mL]; pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

International issues:

Avinza [Puerto Rico] may be confused with Evista brand name for raloxifene [US, Canada, and multiple international markets] and Invanz brand name for ertapenem [US, Canada, Qatar]

Other safety concerns:

MSO<sub>4</sub> and MS are error-prone abbreviations (mistaken as magnesium sulfate)

Use care when prescribing and/or administering morphine solutions. These products are available in different concentrations. Always prescribe dosage in mg; **not** by volume (mL).

Use caution when selecting a morphine formulation for use in neurologic infusion pumps (eg, Medtronic delivery systems). The product should be appropriately labeled as "preservative-free" and suitable for intraspinal use via continuous infusion. In addition, the product should be formulated in a pH range that is compatible with the device operation specifications.

Significant differences exist between oral and IV dosing. Use caution when converting from one route of administration to another.

Metabolism/Transport Effects

**Substrate** of OCT1, P-glycoprotein (Major with inhibitors), P-glycoprotein (Minor with inducers), UGT1A1, UGT1A8, UGT2B7; **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

**Note:** Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Antiplatelet Agents (P2Y12 Inhibitors): Morphine (Systemic) may decrease antiplatelet effects of Antiplatelet Agents (P2Y12 Inhibitors). Morphine (Systemic) may decrease serum concentrations of Antiplatelet Agents (P2Y12 Inhibitors). Management: Consider alternative anti-ischemic/analgesic therapies (eg, beta-blockers, nitroglycerin) in patients with acute coronary syndromes treated with a P2Y12 inhibitor when possible. The risks associated with other opioids are unknown. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase analgesic effects of Opioid Agonists. Cimetidine may increase respiratory depressant effects of Opioid Agonists. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Eluxadoline: Opioid Agonists may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Enzalutamide: May increase active metabolite exposure of Morphine (Systemic). Specifically, morphine-6-glucuronide (M6G) concentrations may be increased. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lomitapide: May increase serum concentrations of P-glycoprotein/ABCB1 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Lonafarnib: May increase serum concentrations of P-glycoprotein/ABCB1 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Lumacaftor and Ivacaftor: May increase serum concentrations of P-glycoprotein/ABCB1 Substrates (High risk with Inhibitors or Inducers). Lumacaftor and Ivacaftor may decrease serum concentrations of P-glycoprotein/ABCB1 Substrates (High risk with Inhibitors or Inducers). *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Morphine (Systemic). *Risk X: Avoid*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid agonists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

P-glycoprotein/ABCB1 Inhibitors: May increase serum concentrations of Morphine (Systemic). *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

RifAMPin: May decrease serum concentrations of Morphine (Systemic). *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Serotonergic Agents (High Risk): Opioid Agonists may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Vimseltinib: May increase serum concentrations of P-glycoprotein/ABCB1 Substrates (High risk with Inhibitors). Management: Avoid concomitant use of vimseltinib and P-gp substrates when possible. If combined, administer vimseltinib at least 4 hours before the P-gp substrate. *Risk D: Consider Therapy Modification*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

#### Food Interactions

Ethanol: Alcoholic beverages or ethanol-containing products may disrupt ER capsule formulations resulting in rapid release of entire morphine dose. Management: Avoid alcohol.

Food: Administration of oral morphine solution with food may increase bioavailability (ie, a report of 34% increase in morphine AUC when morphine oral solution followed a high-fat meal). The bioavailability of ER tablets or capsules does not appear to be affected by food. Management: Take consistently with or without meals.

#### Reproductive Considerations

Chronic opioid use may cause hypogonadism and hyperprolactinemia which may decrease fertility in patients of reproductive potential. Menstrual cycle disorders (including amenorrhea), erectile dysfunction, and impotence have been reported. The incidence of hypogonadism may be increased with the use of opioids in high doses or long-acting opioid formulations. It is not known if the effects on fertility are reversible. Monitor patients on long-term therapy (de Vries 2020; Gadelha 2022).

Consider family planning, contraception, and the effects on fertility prior to prescribing opioids for chronic pain to patients who could become pregnant (ACOG 2017; CDC [Dowell 2022]).

#### Pregnancy Considerations

Morphine crosses the placenta.

Maternal use of opioids may be associated with poor fetal growth, stillbirth, and preterm delivery (CDC [Dowell 2022]). Opioids used as part of obstetric analgesia/anesthesia during labor and delivery may temporarily affect the fetal heart rate (ACOG 2019).

Neonatal abstinence syndrome (NAS)/neonatal opioid withdrawal syndrome (NOWS) may occur following prolonged in utero exposure to opioids (CDC [Dowell 2022]). NAS/NOWS may be life-threatening if not recognized and treated and requires management according to protocols developed by neonatology experts. Presentation of symptoms varies by opioid characteristics (eg, immediate release, sustained release), time of last dose prior to delivery, drug metabolism (maternal, placental, and infant), net placental transfer, as well as other factors (AAP [Hudak 2012]; AAP [Patrick 2020]). Clinical signs characteristic of withdrawal following in utero opioid exposure include excessive crying or easily irritable, fragmented sleep (<2 to 3 hours after feeding), tremors, increased muscle tone, or GI dysfunction (hyperphagia, poor feeding, feeding intolerance, watery or loose stools) (Jilani 2022). NAS/NOWS occurs following chronic opioid exposure and would not be expected following the use of opioids at delivery (AAP [Patrick 2020]).

Monitor infants of mothers on long-term/chronic opioid therapy for symptoms of withdrawal. Symptom onset reflects the half-life of the opioid used. Monitor infants for at least 3 days following exposure to immediate-release opioids; monitor for at least 4 to 7 days following exposure to sustained-release opioids (AAP [Patrick 2020]; CDC [Dowell 2022]). Monitor newborns for excess sedation and respiratory depression when opioids are used during labor.

When opioids are needed to treat acute pain in pregnant patients, the lowest effective dose for only the expected duration of pain should be prescribed (CDC [Dowell 2022]).

Morphine injection is commonly used for the treatment of pain during labor and immediately postpartum (ACOG 2019). Not all dosage forms are appropriate for this use. Opioid use for pain following vaginal or cesarean delivery should be made as part of a shared decision-making process. A step-wise, multimodal approach to managing postpartum pain is recommended. A low-dose, low-potency, short-acting opioid such as morphine can be used to treat acute pain associated with delivery when needed (ACOG 2021).

Opioids are not preferred for the treatment of chronic noncancer pain during pregnancy; consider strategies to minimize or avoid opioid use. Advise pregnant patients requiring long-term opioid use of the risk of NAS/NOWS and provide appropriate treatment for the neonate after delivery. NAS/NOWS is an expected and treatable condition following chronic opioid use during pregnancy and should not be the only reason to avoid treating pain with an opioid in pregnant patients (ACOG 2017; CDC [Dowell 2022]). Do not abruptly discontinue opioids during pregnancy; taper prior to discontinuation when appropriate, considering the risks to the pregnant patient and fetus if maternal withdrawal occurs (CDC [Dowell 2022]).

#### Breastfeeding Considerations

Morphine is present in breast milk. M6G, the active metabolite of morphine, can also be detected in breast milk (Baka 2002).

Reported concentrations of morphine in breast milk are variable.

Morphine has been detected in the urine of a breastfeeding infant following maternal epidural dosing after cesarean delivery (Zakowski 1993) and the serum of an infant following chronic maternal oral use (Robieux 1990).

Nonopioid analgesics are preferred for lactating patients who require pain control peripartum or for surgery outside of the postpartum period. When opioids are needed for lactating patients, use the lowest effective dose for the shortest duration of time to limit adverse events in the mother and breastfeeding infant. When an opioid is needed to treat maternal pain, morphine is one of the preferred agents. Analgesics delivered by patient-controlled analgesia or administered by the epidural route help limit infant exposure (AAP [Sachs 2013]; ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; WHO 2002).

When chronic opioids are prescribed prenatally and continued postpartum, breastfeeding may be initiated to help mitigate potential newborn withdrawal; monitor both the mother and the infant (AAP [Meek 2022]; AAP [Patrick 2020]).

Monitor infants exposed to opioids via breast milk for drowsiness, sedation, feeding difficulties, or limpness (ACOG 2019). Withdrawal symptoms may occur when maternal use is discontinued, or

breastfeeding is stopped.

#### Dietary Considerations

Morphine may cause GI upset; take with food if GI upset occurs. Be consistent when taking morphine with or without meals.

#### Monitoring Parameters

Pain control, respiratory and mental status; BP; signs of misuse, abuse, and substance use disorder; signs or symptoms of hypogonadism or hypoadrenalism (Brennan 2013); signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [non-cardiac], dysphagia, regurgitation).

Alternate recommendations: Subacute or chronic pain (long-term therapy outside of end-of-life or palliative care, active cancer treatment, sickle cell disease, or medication-based opioid use disorder treatment): Evaluate benefits/risks of opioid therapy within 1 to 4 weeks of treatment initiation and with dose increases. In patients with subacute pain initially treated for acute pain, reassess pain and function after 30 days to address potentially reversible causes of pain and prevent unintentional long-term opioid therapy. In patients on long-term therapy, reevaluate benefits/risks every 3 months during therapy or more frequently in patients at increased risk of overdose or opioid use disorder. Toxicology testing is recommended prior to initiation and at least yearly (includes controlled prescription medications, illicit drugs of abuse, and benzodiazepines). State prescription drug monitoring program data should be reviewed by clinicians prior to initiation and periodically during therapy (frequency ranging from every prescription to every 3 months) (CDC [Dowell 2022]).

Critically ill: The Numeric Rating Scale should be used in patients who are able to self-report pain. In patients who are unable to self-report pain, the Behavioral Pain Scale and the Critical-Care Pain Observational Tool can be used in intubated or nonintubated patients (SCCM [Devlin 2018]).

Epidural/intrathecal: Patients should be observed in a fully equipped and staffed environment for at least 24 hours following initiation, and as appropriate for the first several days after catheter implantation. Opioid reversal agents (eg, naloxone, nalmeferene) should be immediately available. Patient should remain in this environment for at least 24 hours following the initial dose. For patients receiving Infumorph or Mitigo via microinfusion device, patient may be observed, as appropriate, for the first several days after catheter implantation.

**Alternative monitoring recommendations** (Bujedo 2012a): Epidural: **Note:** All patients receiving neuraxial opioids should be monitored for adequate ventilation (eg, respiratory rate, depth of respiration [without disturbing patient]), oxygenation (eg, pulse oximetry when appropriate), and level of consciousness.

*Single dose:* Monitor patient for a minimum of 24 hours after administration with a frequency of at least once per hour for the first 12 hours after administration, followed by at least once every 2 hours for the next 12 hours (ie, from 12 to 24 hours after administration). After 24 hours, frequency dictated by overall clinical condition and concurrent medications.

*Continuous infusion or patient controlled epidural analgesia (PCEA):* Monitor patient during the entire duration of the infusion with a frequency of at least once per hour for the first 12 hours, followed by at least once every 2 hours for the next 12 hours (ie, from 12 to 24 hours after administration). After 24 hours, monitor at least once every 4 hours. After discontinuation of infusion or PCEA, frequency dictated by overall clinical condition and concurrent medications.

**Note:** Also refer to institution specific protocols as appropriate.

#### Mechanism of Action

Binds to opioid receptors in the CNS, causing inhibition of ascending pain pathways, altering the perception of and response to pain; produces generalized CNS depression

#### Pharmacokinetics (Adult Data Unless Noted)

Onset of action (patient dependent; dosing must be individualized): Oral (immediate release): ~30 minutes; IV: 5 to 10 minutes.

Duration (patient dependent; dosing must be individualized): Pain relief:

Immediate-release formulations (tablet, oral solution, injection): 3 to 5 hours.

Extended-release capsule and tablet: 8 to 24 hours (formulation dependent).

Epidural or intrathecal: Single dose: Up to 24 hours (Bujedo 2012a).

Suppository: 3 to 7 hours.

Absorption: Variable.

Distribution: Distributes to skeletal muscle, liver, kidneys, lungs, intestinal tract, spleen and brain;  
 $V_d$ :

Preterm neonates:  $2.04 \pm 1.01$  L/kg (range: 0.6 to 3.68 L/kg) (Chay 1992).

Term neonates:  $2.07 \pm 1.26$  L/kg (range: 0.24 to 3.34 L/kg) (Chay 1992).

Children with cancer (age: 1.7 to 18.7 years): Median: 5.2 L/kg; a significantly higher  $V_d$  was observed in children <11 years (median: 7.1 L/kg) versus >11 years (median: 4.7 L/kg) (Hunt 1999).

Adults: 1 to 6 L/kg; binds to opioid receptors in the CNS and periphery (eg, GI tract).

Protein binding: Premature infants: 18% to 22% (Bhat 1990); Adults: 20% to 35%.

Metabolism: Hepatic via conjugation with glucuronic acid primarily to morphine-6-glucuronide (M6G) (active analgesic) morphine-3-glucuronide (M3G) (inactive as analgesic; may contribute to CNS stimulation [Lugo 2002]); minor metabolites include morphine-3-6-diglucuronide; other minor metabolites include normorphine (active) and morphine 3-ethereal sulfate. Due to immature hepatic enzyme development, neonates exhibit at least a 50% lower glucuronidation rate than older children (Cravero 2019).

Bioavailability: Oral: 17% to 33% (first-pass effect limits oral bioavailability); Nebulization:  $5.5 \pm 3.2\%$  (Masood 1996).

Half-life elimination:

Pediatric: **Note:** Values listed are based on IV dosing (IV bolus and/or continuous IV infusion):

Preterm neonates:  $10.6 \pm 2.7$  hours (range: 5.5 to 13.3 hours) (Chay 1992).

Term neonates:  $7.6 \pm 2.6$  hours (range: 4.5 to 11 hours) (Chay 1992).

Infants 1 to 3 months: Median: 6.2 hours (range: 5 to 9.9 hours) (McRorie 1992).

Infants 3 to 6 months: Median: 4.5 hours (range: 3.8 to 7.3 hours) (McRorie 1992).

Infants 6 months to Children 2.5 years: Median: 2.9 hours (range: 1.4 to 7.8 hours) (McRorie 1992).

Preschool children: 1 to 2 hours (Oikkola 1995).

Children with sickle cell disease (age: 6 to 19 years):  $\sim 1.3$  hours (Dampier 1995).

Adults:

Extended release:

Capsules (once- or twice-daily formulation [polymer coated pellets]): Terminal: 11 to 13 hours.

Capsules (once-daily formulation [pellets]): Terminal:  $\sim 24$  hours.

Tablets: Not reported.

Immediate-release forms: 2 to 4 hours.

Time to peak:

Plasma:

Capsules, extended release (once- or twice-daily formulation [polymer coated pellets]):  $\sim 10$  hours.

Capsules, extended release (once-daily formulation [pellets]): Capsules consist of an IR and an ER component. Following a single dose of 60 mg, morphine concentrations were achieved within 30 minutes and maintained for the 24-hour dosing interval.

Tablets, extended release: 3 to 4 hours.

Tablets (immediate release), oral solution, epidural: 1 hour.

Suppository: 20 to 60 minutes.

SUBQ: 50 to 90 minutes.

IM: 30 to 60 minutes.

IV: 20 minutes.

Cerebrospinal fluid: After an oral dose of controlled release morphine concentrations peak at 8 hours for both normal and reduced renal function; morphine-6-glucuronide (active analgesic) and morphine-3-glucuronide distribution into the CNS may be delayed peaking at 12 hours in patients with normal renal function or up to 24 hours in patients with ESRD (peak level of morphine-6-glucuronide is ~15 times higher than patients with normal renal function) (D'Honneur 1994).

Excretion: Urine (primarily as morphine-3-glucuronide, neonates: 3% to 15%; adults: ~2% to 12% excreted unchanged); feces (~7% to 10%). It has been suggested that accumulation of morphine-6-glucuronide might cause toxicity with renal insufficiency. All of the metabolites (ie, morphine-3-glucuronide, morphine-6-glucuronide, and normorphine) have been suggested as possible causes of neurotoxicity (eg, myoclonus).

Clearance: **Note:** In pediatric patients, adult values are reached by 6 months to 2.5 years of age (McRorie 1992).

Preterm: 0.5 to 3 mL/minute/kg (Olkola 1995).

Neonates 1 to 7 days: Median: 5.5 mL/minute/kg (range: 3.2 to 8.4 mL/minute/kg) (McRorie 1992).

Neonates 8 to 30 days: Median: 7.4 mL/minute/kg (range: 3.4 to 13.8 mL/minute/kg) (McRorie 1992).

Infants 1 to 3 months: Median: 10.5 mL/minute/kg (range: 9.8 to 20.1 mL/minute/kg) (McRorie 1992).

Infants 3 to 6 months: Median: 13.9 mL/minute/kg (range: 8.3 to 24.1 mL/minute/kg) (McRorie 1992).

Infants 6 months to Children 2.5 years: Median: 21.7 mL/minute/kg (range: 5.8 to 28.6 mL/minute/kg) (McRorie 1992).

Preschool children: 20 to 40 mL/minute/kg (Olkola 1995).

Children with cancer (age: 1.7 to 18.7 years): Median: 23.1 mL/minute/kg; a significantly higher clearance was observed in children <11 years (median: 37.4 mL/minute/kg) versus >11 years (median: 21.9 mL/minute/kg) (Hunt 1999).

Children with sickle cell disease (age: 6 to 19 years): ~36 mL/minute/kg (range: 6 to 59 mL/minute/kg) (Dampier 1995).

Adults: 20 to 30 mL/minute/kg.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Pediatric: Increased pharmacokinetic variability in neonates likely due to immaturity of hepatic glucuronidation, alterations in hepatic blood flow during acute illness, and blood shunting away from the liver by the ductus venosus (Bhat 1990).

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

MS Contin: [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2025/019516s062lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/019516s062lbl.pdf)

Oral solution: [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2023/022195s021lbl.pdf#page=23](https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/022195s021lbl.pdf#page=23)

Oral tablet: [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2023/022207s011lbl.pdf#page=29](https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/022207s011lbl.pdf#page=29)

#### Patient Counseling Points

(For additional information see "Morphine: Patient drug information")

#### **What is this drug used for?**

- It is used to manage pain when non-opioid pain drugs do not treat your pain well enough or you cannot take them.
- Some products are used to manage pain only when daily pain treatment is needed for a long time.

**All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:**

- Feeling dizzy, sleepy, tired, or weak
- Dry mouth
- Constipation, diarrhea, stomach pain, upset stomach, throwing up, or decreased appetite
- Headache
- Anxiety
- Sweating a lot

**WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:**

#### **All products:**

- Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating
- Severe dizziness or passing out
- Chest pain or pressure or a fast heartbeat
- Feeling confused
- Trouble breathing, slow breathing, or shallow breathing
- Noisy breathing
- Breathing problems during sleep (sleep apnea)
- Seizures
- Severe bowel problem like severe constipation or stomach pain
- Depression or other mood changes
- Swelling in the arms or legs
- Fever, chills, or sore throat
- Pain when passing urine
- A burning, numbness, or tingling feeling that is not normal
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache
- Severe adrenal gland problem like if you feel very tired or weak, you pass out, or you have severe dizziness, very upset stomach, throwing up, or decreased appetite

- A lowered interest in sex, fertility problems, no menstrual period, or ejaculation problems
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

### **Injection (if given into the spine):**

- Not able to move
- Muscle spasm
- Trouble controlling body movements
- Trouble passing urine

**Note:** This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

**Consumer Information Use and Disclaimer:** This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Morphgesic sr | Sevredol;
- (AR) Argentina: Algedol | M eslon | Morfina | Mst continus;
- (AT) Austria: Kapanol csr | M Dolor | M-long | Muididol | Substitol;
- (AU) Australia: Anamorph | Apotex Morphine MR | Kapanol | Momex | Morphine mr an | Morphine sulphate | Ms | Ms contin | Sevredol;
- (BD) Bangladesh: Morphine sulphate | Msl;
- (BE) Belgium: Kapaject | Kapanol | Morphine teva generics belgium | Morphisin | Ms contin | Ms direct | Skenan;
- (BG) Bulgaria: Doltard | Morphine sulphate | Mst continus | Sevredol | Skenan | Slovalgin | Substitol;
- (BR) Brazil: Dimorf | Dimorf lc | Dolo moff | Ms | Sulfato de morfina;
- (CH) Switzerland: Morphine Bioren | Morphinsulfat Amino | Mst continus | Sevredol;
- (CL) Chile: M eslon;
- (CN) China: Mei shi kang ding | Morphine | Ms contin;
- (CR) Costa Rica: Morfina Sulfato;
- (CZ) Czech Republic: M eslon Sr | Mst continus | Mst uno | Sevredol | Skenan | Slovalgin;
- (DE) Germany: Capros | Capros akut | Kapanol | M Dolor | M-beta | M-long | Mogetic | Morixon | Morph | Morph 1 a pharma | Morph Sandoz | Morphanton | Morphin | Morphin al | Morphin aristo | Morphin AWD | Morphin neuraxpharm | Morphin painbreak | Morphinsulfat Abz | Morphinsulfat Glenmark | Morphinsulfat GRY | Morphinsulfat ps pharma | Morphinsulfat-CT | Msr | Mst | Mst continus | Onkomorphin | Sevredol | Substitol;
- (DK) Denmark: Contalgin uno | Malfin;
- (EC) Ecuador: M eslon | Morfina;

- (EE) Estonia: Doltard | Ms contin | Mst continus | Mxl | Sevredol;
- (EG) Egypt: Sevredol;
- (ES) Spain: Mst continus | Mst unicontinus | Sevredol | Skenan | Zomorph;
- (ET) Ethiopia: Vermor;
- (FI) Finland: Dolcontin | Duralgin | Kapanol;
- (FR) France: Actiskenan | Kapanol | Moscontin | Sevredol | Skenan | Skenan lp;
- (GB) United Kingdom: Depodur | Filnarine | Morcap | Morphgesic | Morphine sulphate | Mst continus | Mxl | Rhotard | Sevredol | Srm rhotard | Zomorph;
- (GR) Greece: Mongol | Morficontin;
- (HK) Hong Kong: Morphine sulphate | Mst continus;
- (HR) Croatia: Mst continus | Sevredol;
- (HU) Hungary: M eslon | Moretal | Mst continus | Mst uno | Sevredol;
- (ID) Indonesia: Mst continus;
- (IE) Ireland: Morstel | Mst continus | Mxl | Sevredol | Slo morph;
- (IL) Israel: Mcr | Mir | Morphex CR | Msp;
- (IN) India: Morcon | Morcontin | Morf | Morphitroy | Rumorf | Rumorf cr;
- (IT) Italy: Ms contin | Skenan | Twice;
- (JP) Japan: Kadian | Kadian sankyo | Kaadian shionogi | Kadian takeda | Kadian tanabe | Morphes | Morphine sulfate hydrate | Ms contin | Ms contin dainippo | Ms contin sankyo | Ms contin takeda | Ms contin tanabe | Ms twicelon | P guard;
- (KE) Kenya: Mst continus | Pharma Q Morphine | Rumorf | Vermor | Zomorph;
- (KR) Korea, Republic of: Contimor | Emstin | Esmolpin | Hana morphine sulfate | Longphine | M.p.s | M.S.R | Mps | Ms contin | MSR SR | Opidul | S-morphine;
- (LB) Lebanon: Mst continus;
- (LT) Lithuania: Doltard | Kapanol | Ms contin | Mst contin | Sevredol;
- (LU) Luxembourg: Kapanol | Ms contin | Ms direct | Skenan;
- (LV) Latvia: Doltard | Mst continus | Mxl | Sevredol;
- (MA) Morocco: Moscontin;
- (MX) Mexico: Analfin | Analfin Lc | Graten | Graten lc | Graten li | Morfina | Mst continus f;
- (MY) Malaysia: Kapanol | Mst continus | Srm rhotard | Statex;
- (NL) Netherlands: Kapanol | Morfinesulfaat | Morfinesulfaat Alpharma | Morfinesulfaat aurobindo retard | Morfinesulfaat PCH | Morfinesulfaat sandoz | Ms contin | Noceptin | Sevredol;
- (NO) Norway: Contalgin uno | Dolcontin | Kapanol | Malfin | Morfin | Morfinsulfat | Mst continus;
- (NZ) New Zealand: Arrow Morphine LA | Kapanol | La morph | M eslon | Morphine sulphate | Sevredol;
- (PA) Panama: Morfia;
- (PH) Philippines: Hizon morphine sulfate | Mst continus | Relimal;
- (PK) Pakistan: Magnus | Morphine | Qonza;
- (PL) Poland: Doltard | M eslon | Mst continus | Sevredol | Skenan | Slovalgin | Vendal;
- (PR) Puerto Rico: Arymo er | Avinza | Infumorph | Kaadian | Mitigo | Morphine | Msir | Roxanol;
- (PT) Portugal: Morfina | Morfina Sulfato | Mst | Sevredol | Skenan;
- (QA) Qatar: Morphgesic SR | Oramorph | Sevredol | Zomorph;
- (RO) Romania: Mst continus;
- (RU) Russian Federation: Mst continus;
- (SA) Saudi Arabia: Mst continus | Sevredol | Zomorph;
- (SE) Sweden: Dolcontin | Dolcontin Unotard | Kapanol;
- (SG) Singapore: Mst continus | Srm rhotard | Statex;
- (SI) Slovenia: Kapanol | Mst continus | Sevredol | Substitol;
- (SK) Slovakia: Mst continus | Sevredol | Skenan | Slovalgin;
- (SR) Suriname: Morfinesulfaat | Ms contin;
- (SV) El Salvador: Sulfato de morfina;
- (TH) Thailand: Kapanol | Mst continus;
- (TN) Tunisia: Moscontin | Skenan;
- (TR) Turkey: M eslon | Morfia | Morphine | Mst continus;
- (TW) Taiwan: Mst continus | Mxl;
- (UA) Ukraine: Morphin | Morphine sulphate;

- (UY) Uruguay: Algedol | Dimorf | S.m. | Serenal;
- (ZA) South Africa: Mst continus | Srm rhotard;
- (ZW) Zimbabwe: Morphine sulphate | Morphine sulphate fresenius pf | Pharma Q Morphine

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